QUT Centre for Robotics



ENN583 – Foundations of Robotic Vision

Week 12 – Sem 2 – 2026

Prof Niko Suenderhauf

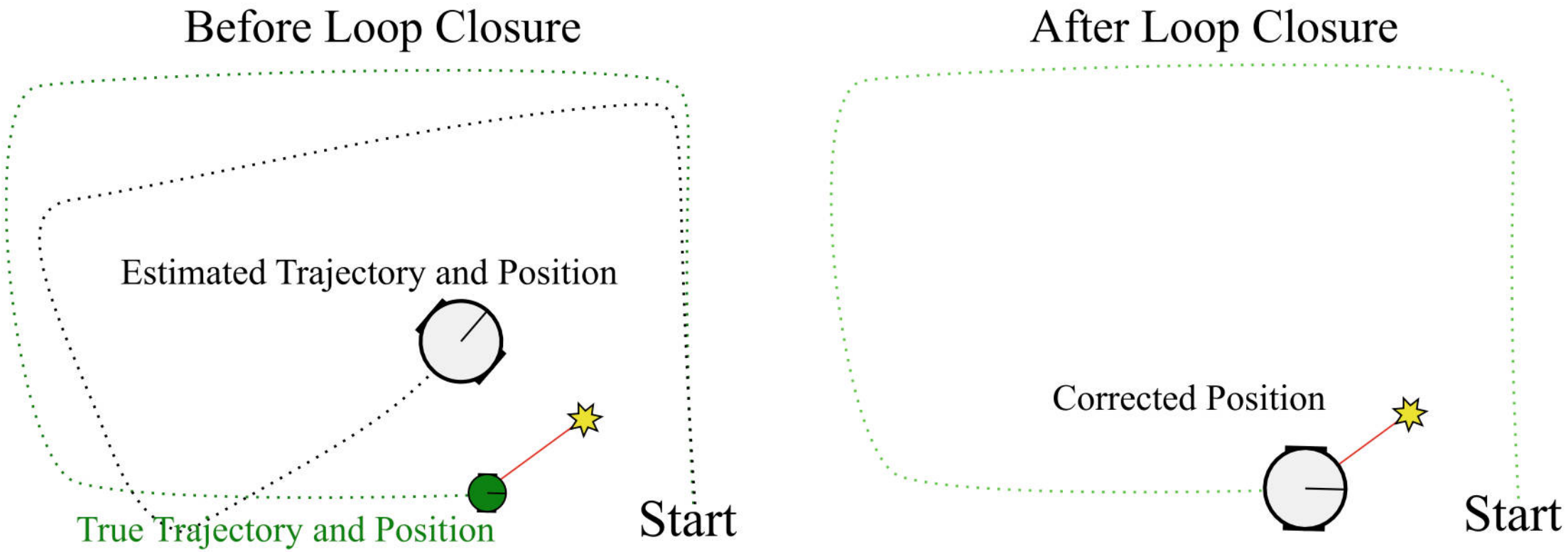
with material by Dr Dimity Miller

Welcome to Week 12!

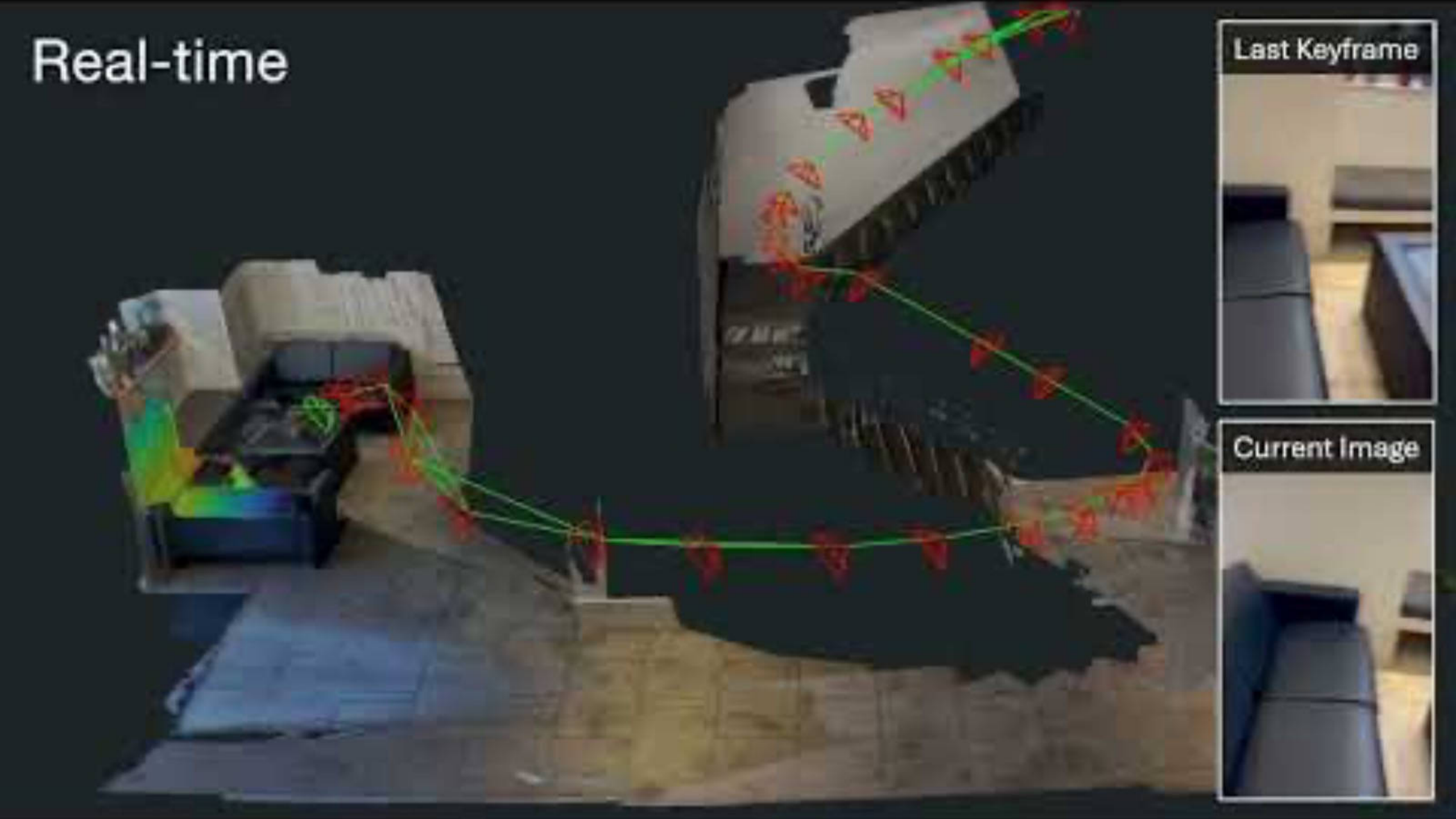


CRICOS No.00213J

Motivation – Remember Visual SLAM?

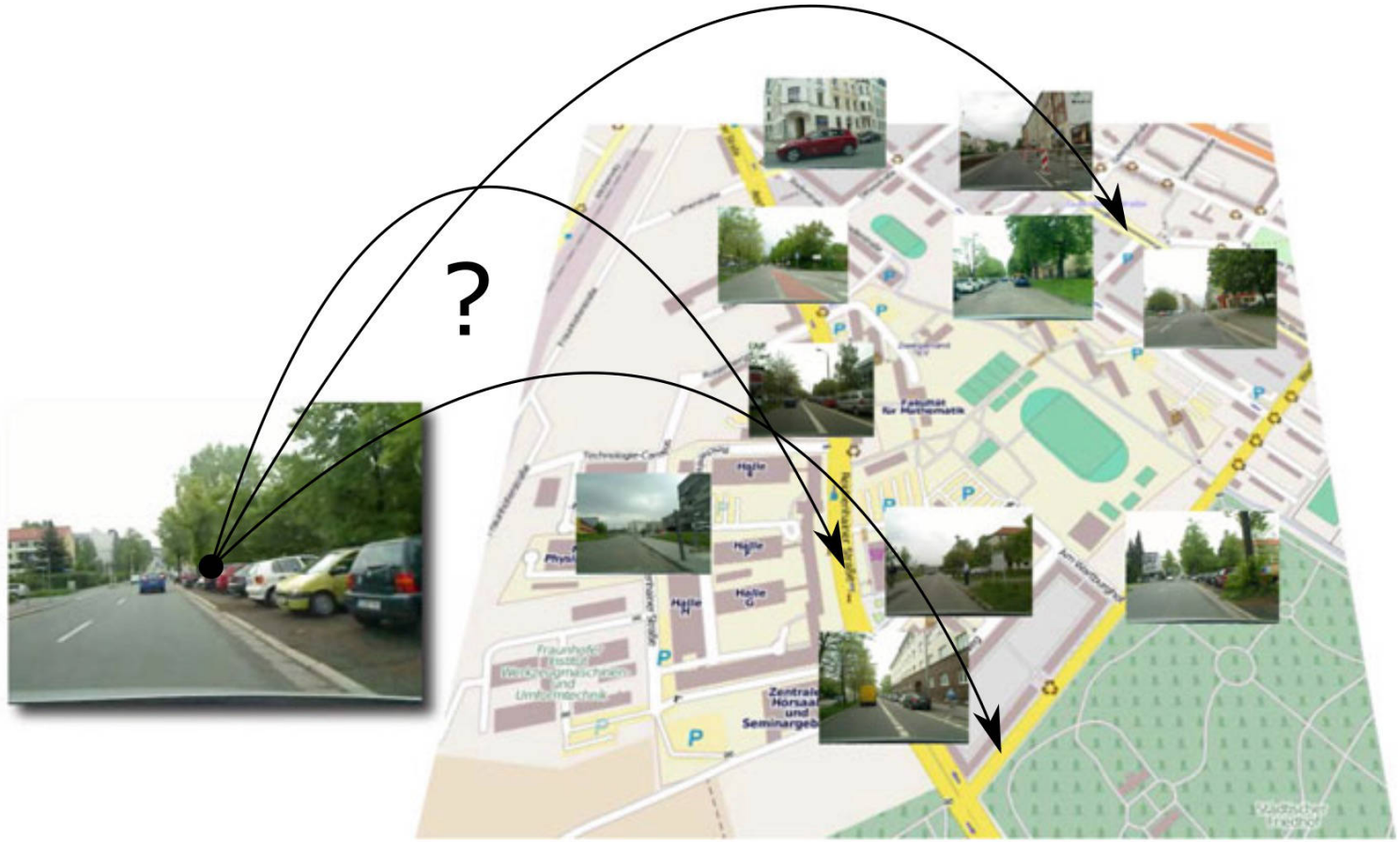


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Real-time
Last Keyframe
Current Image

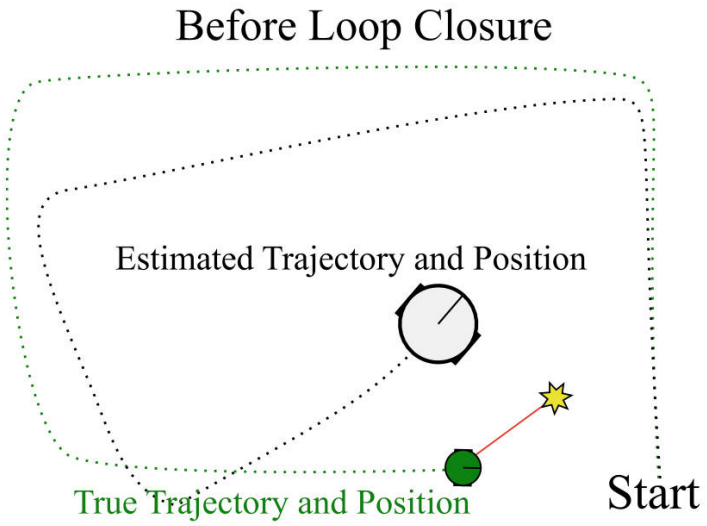
Visual Place Recognition

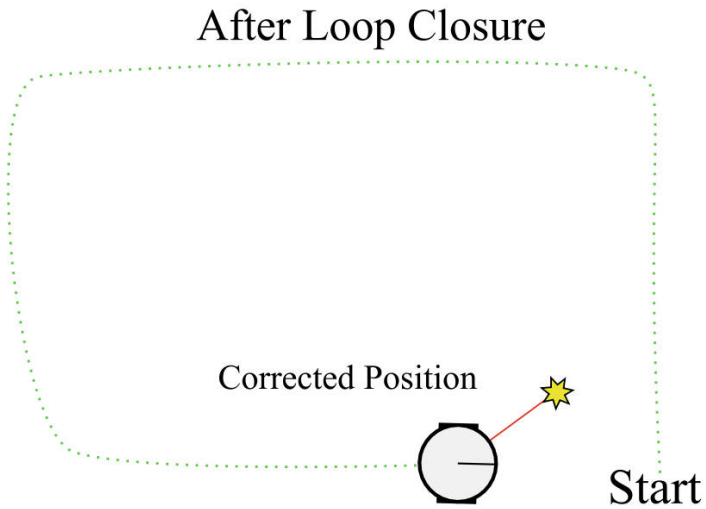
Given an image of a place, how can a robot decide if it has seen this place before?



Why Visual Place Recognition?

- Loop Closure Detection
- Kidnapped Robot Problem
- Initial candidates for full 6DOF localization (PnP)



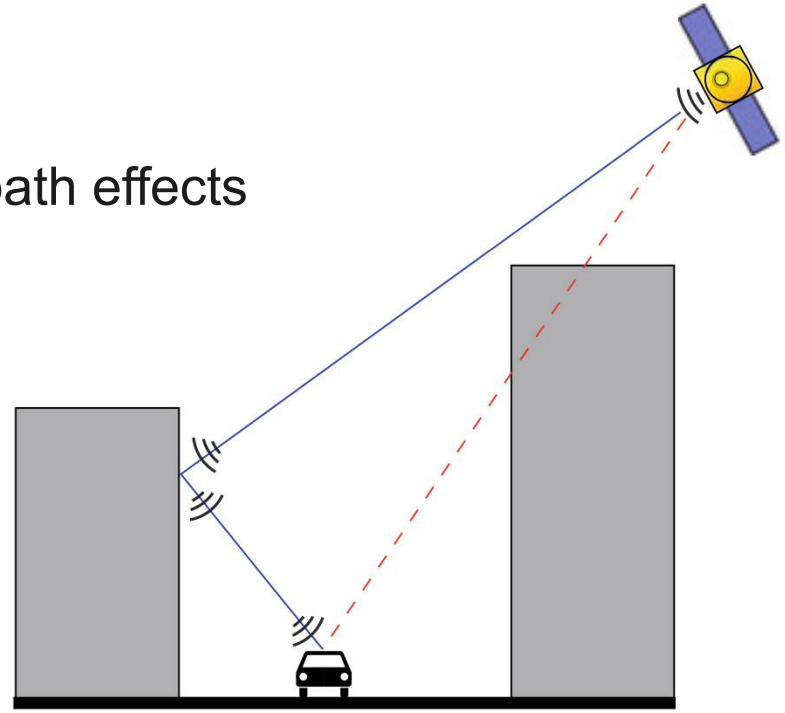


Switchable Constraints for Robust Simultaneous Localization and Mapping and Satellite-Based Localization. Niko Suenderhauf, Springer Nature, 2023.

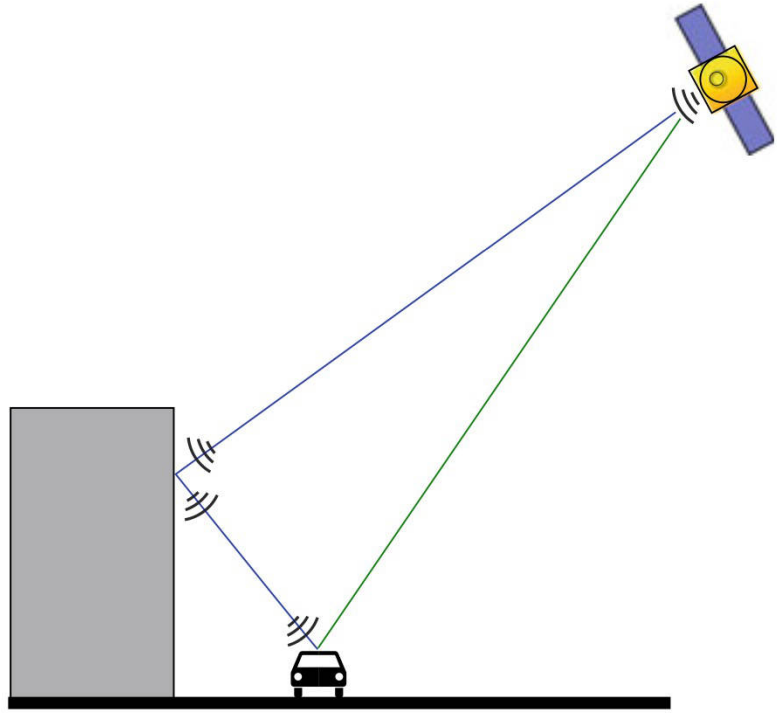
Switchable Constraints for Robust Simultaneous Localization and Mapping and Satellite-Based Localization. Niko Suenderhauf, Springer Nature, 2023.

Why not just rely on GNSS?

- GPS-denied environments
  - Indoor, underground
- Diminished GPS accuracy
  - Non-line-of-sight and multipath effects
  - ”urban canyon”
  - Under canopy
  - Other natural structures
- Desired localisation accuracy



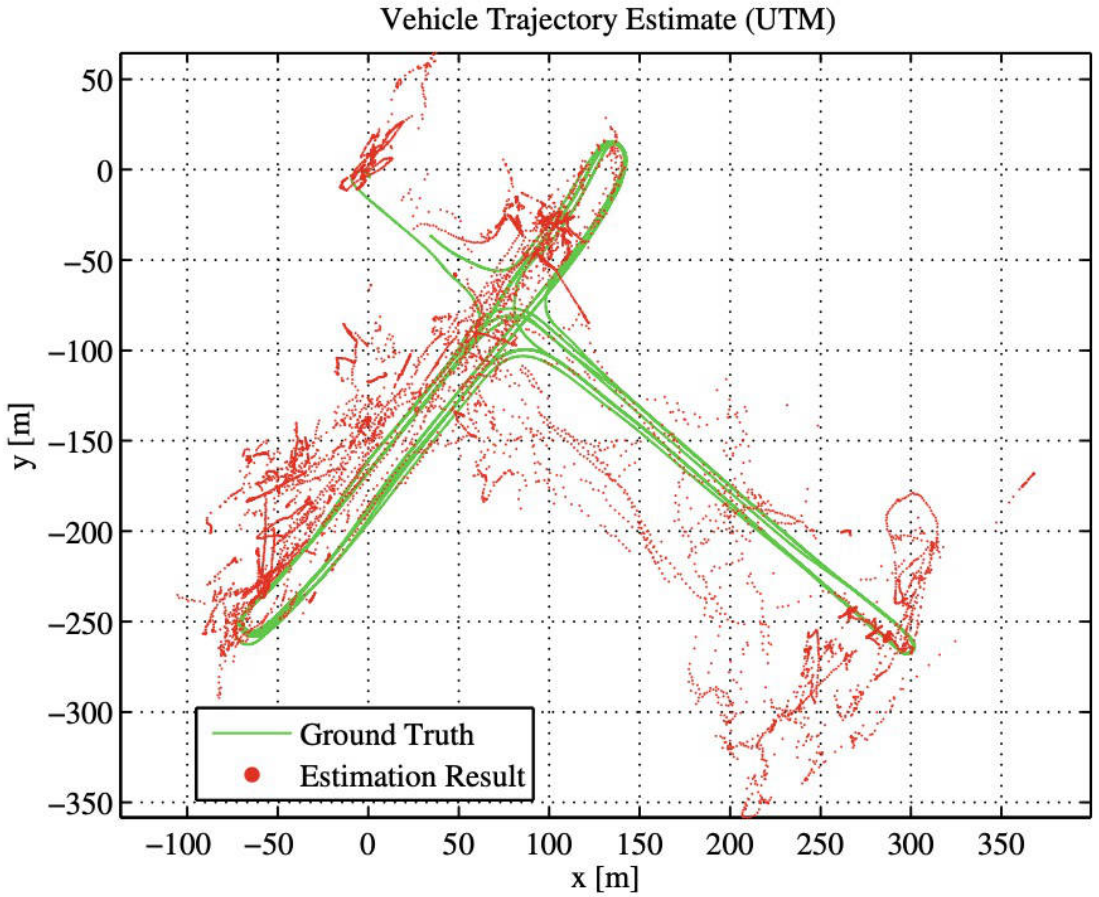
Non-line-of-sight Signals

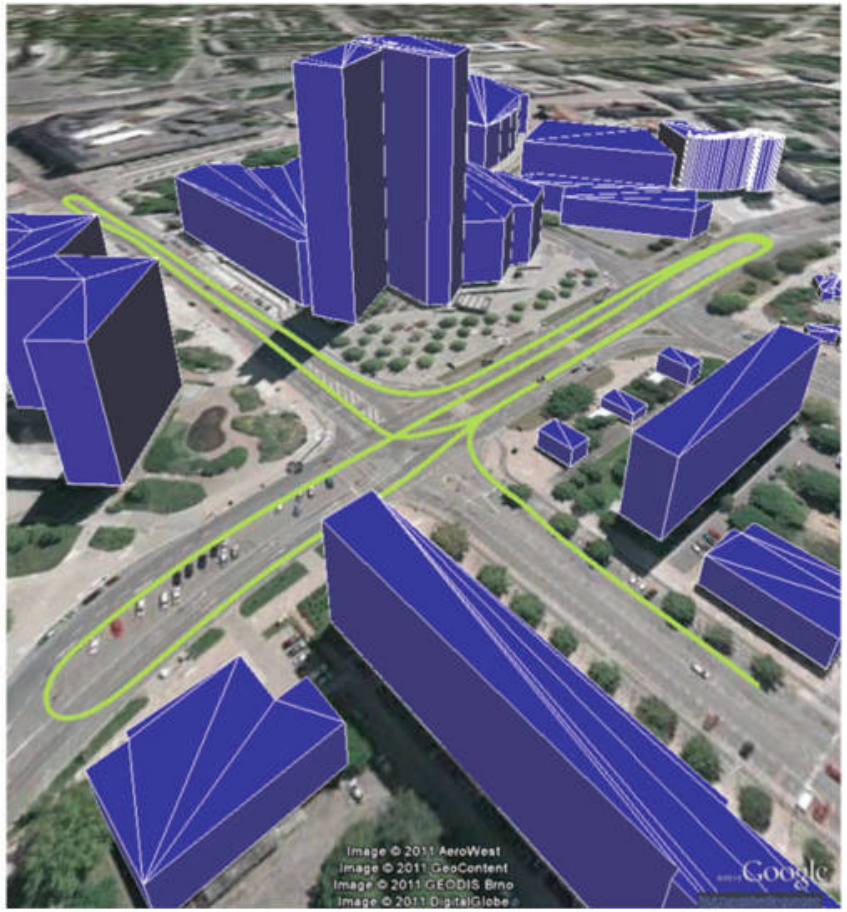


Multipath Signals

Switchable Constraints for Robust Simultaneous Localization and Mapping and Satellite-Based Localization. Niko Suenderhauf, Springer Nature, 2023.

Why not just rely on GNSS?

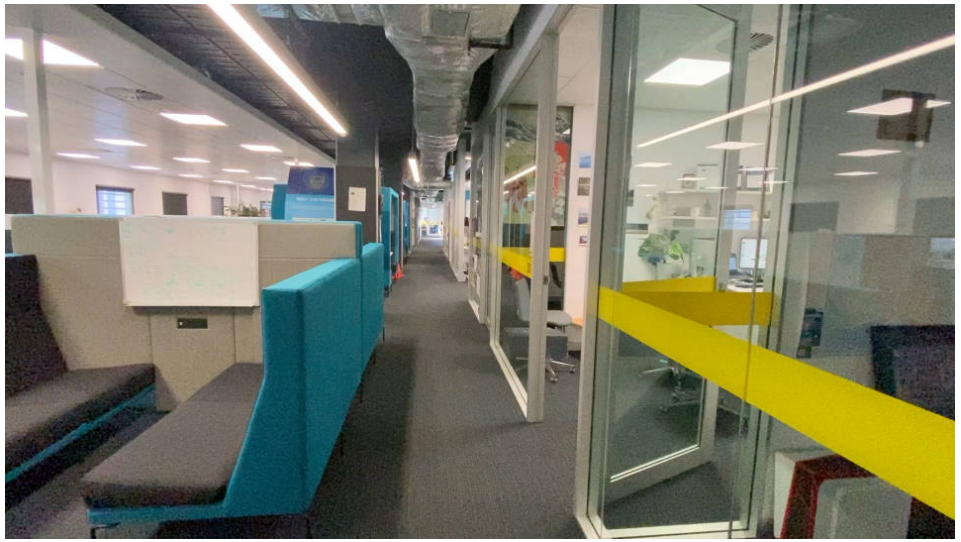


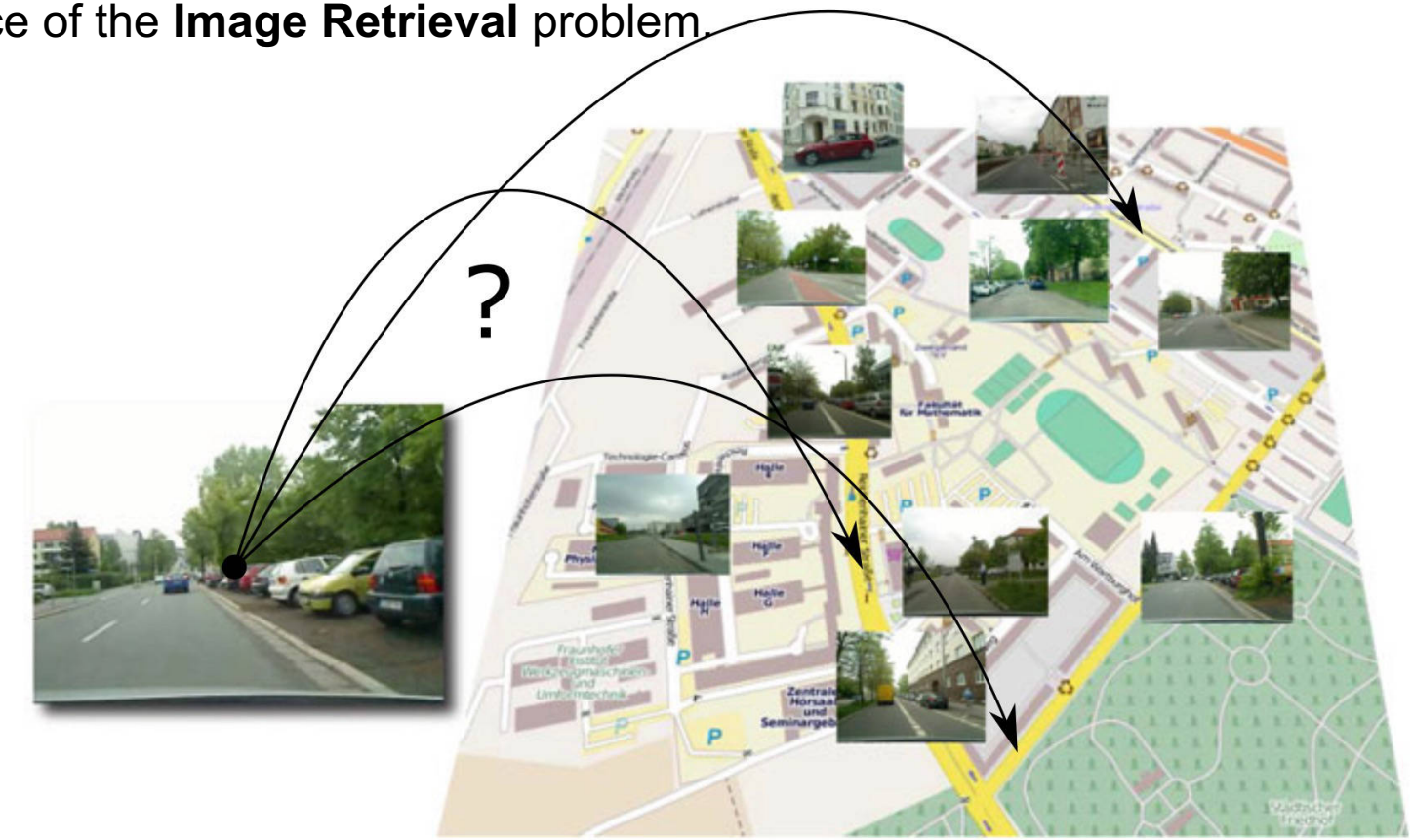


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Visual Place Recognition

Given an image of a place, can a robot decide if it has seen this place before?
This is an instance of the **Image Retrieval** problem.





What is Image Retrieval?

Query image



Database



Database matches



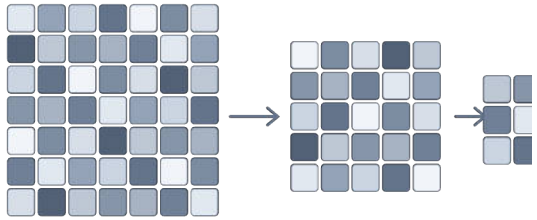
Given a query image, find the most similar/relevant images in a database.

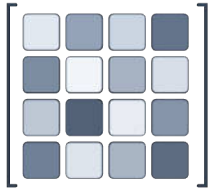
In visual place recognition, “similar” ideally means showing the same physical place, even if appearance or viewpoint has changed.

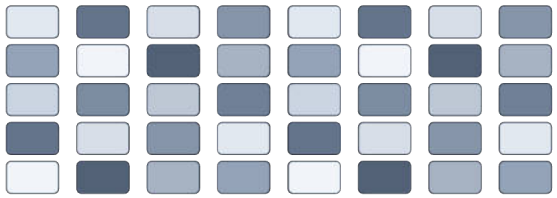
Similarity is measured by comparing the **feature representation** of the images.

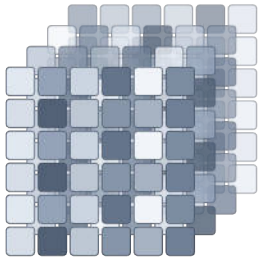
Representation Learning

Now learning a good representation is the task!









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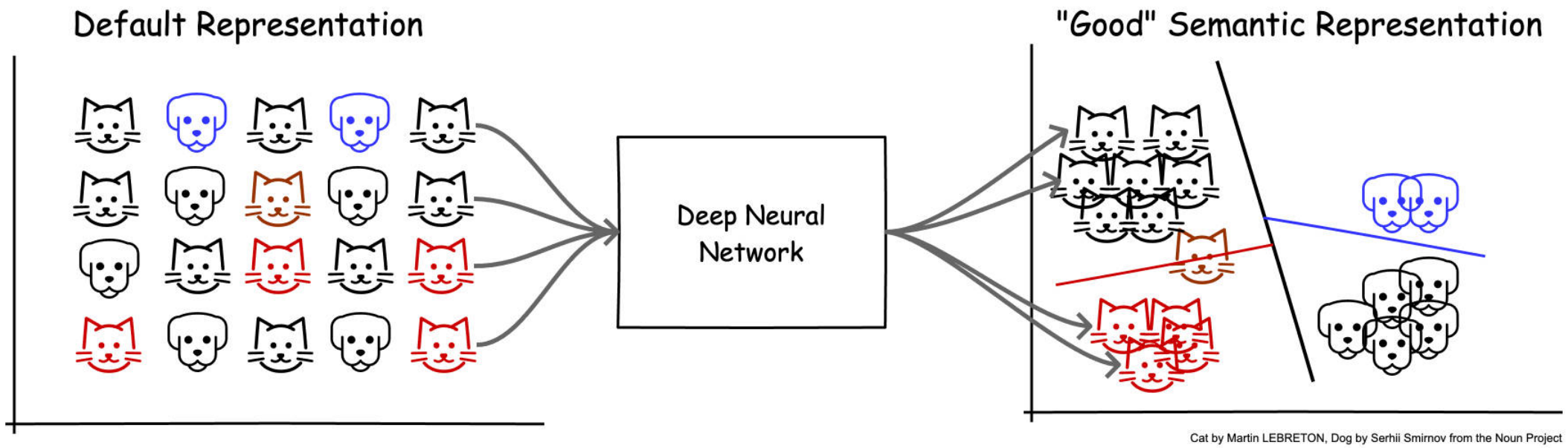


Encoder
"Backbone"



Prediction
Head

What is Representation Learning?

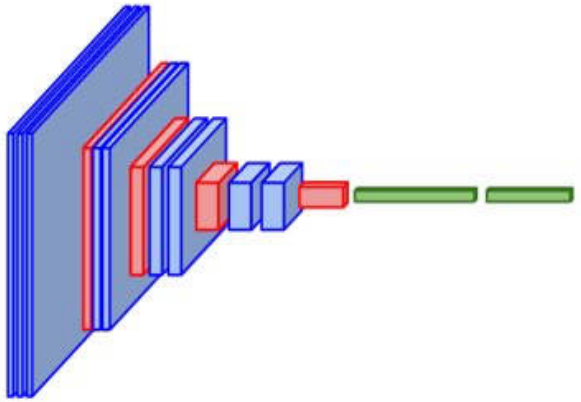


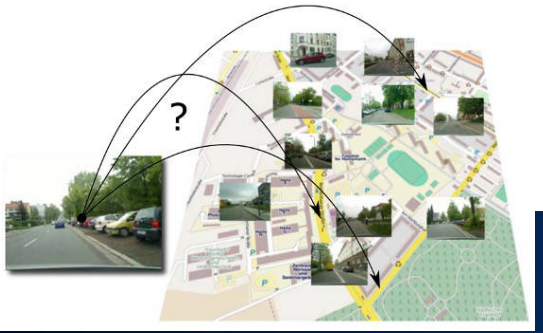
https://blog.fastforwardlabs.com/2020/11/15/representation-learning-101-for-software-engineers.html

VPR as an Image Retrieval Problem

Why not formulate this as a classification task with supervised learning?







Place 1

Place 2

Place 3

Place 4

Place 5

...

Place n

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VPR as an Image Retrieval Problem

Why not formulate this as a classification task with supervised learning?

Retrieval

- Stores examples directly in a database
- New places can be added by inserting new reference images
- Naturally returns the most similar candidates
- Better suited to very large and continually changing sets of places

Classification

- Requires a fixed set of place classes defined during training
- Each new place may require adding/retraining a class
- Produces a class label, not necessarily a useful similarity ranking

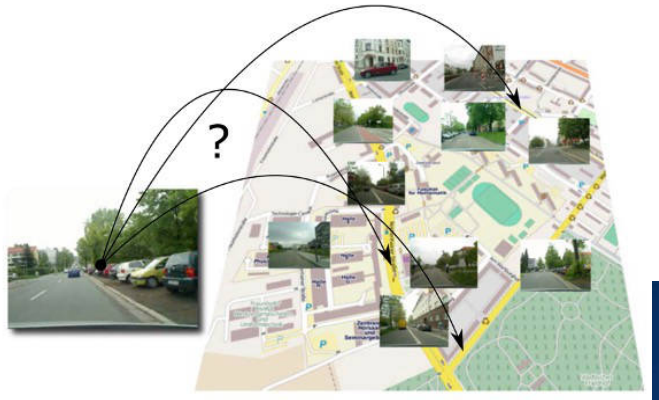
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VPR as an Image Retrieval Problem

Why not formulate this as a classification task with supervised learning?







Nearby places may not form clean, discrete classes; retrieval naturally handles graded similarity between observations.

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VPR as an Image Retrieval Problem

Why not formulate this as a classification task with supervised learning?

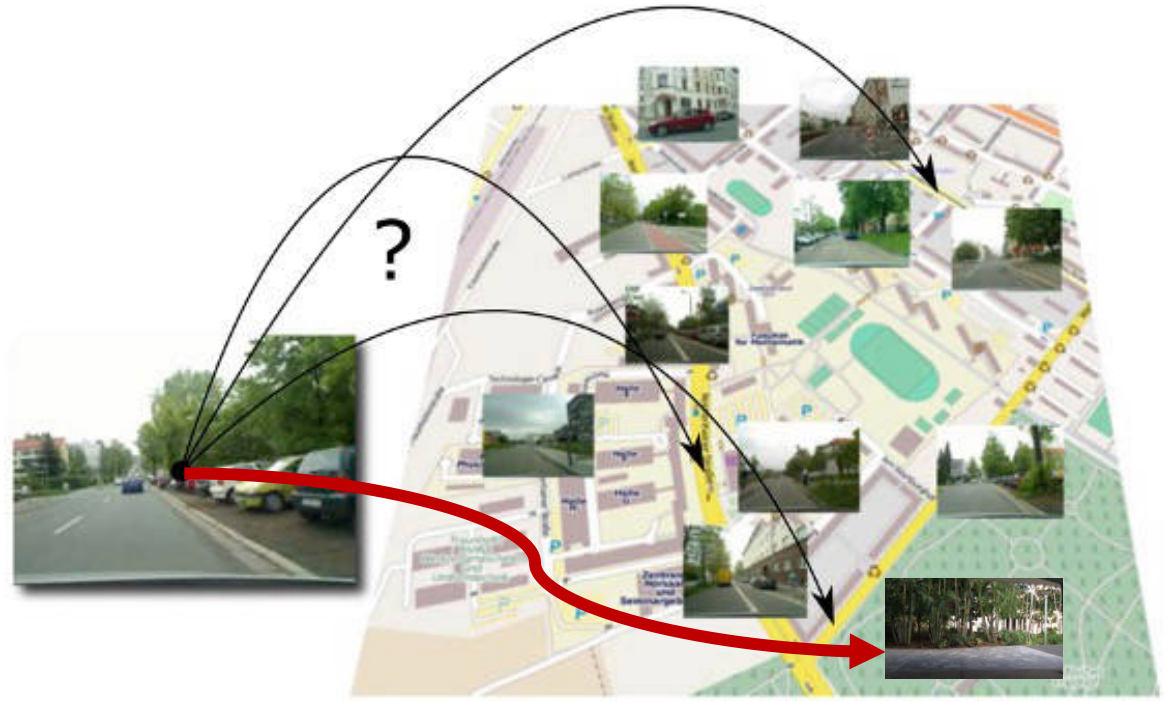


Image Retrieval allows **flexibility,** as it can easily be updated.

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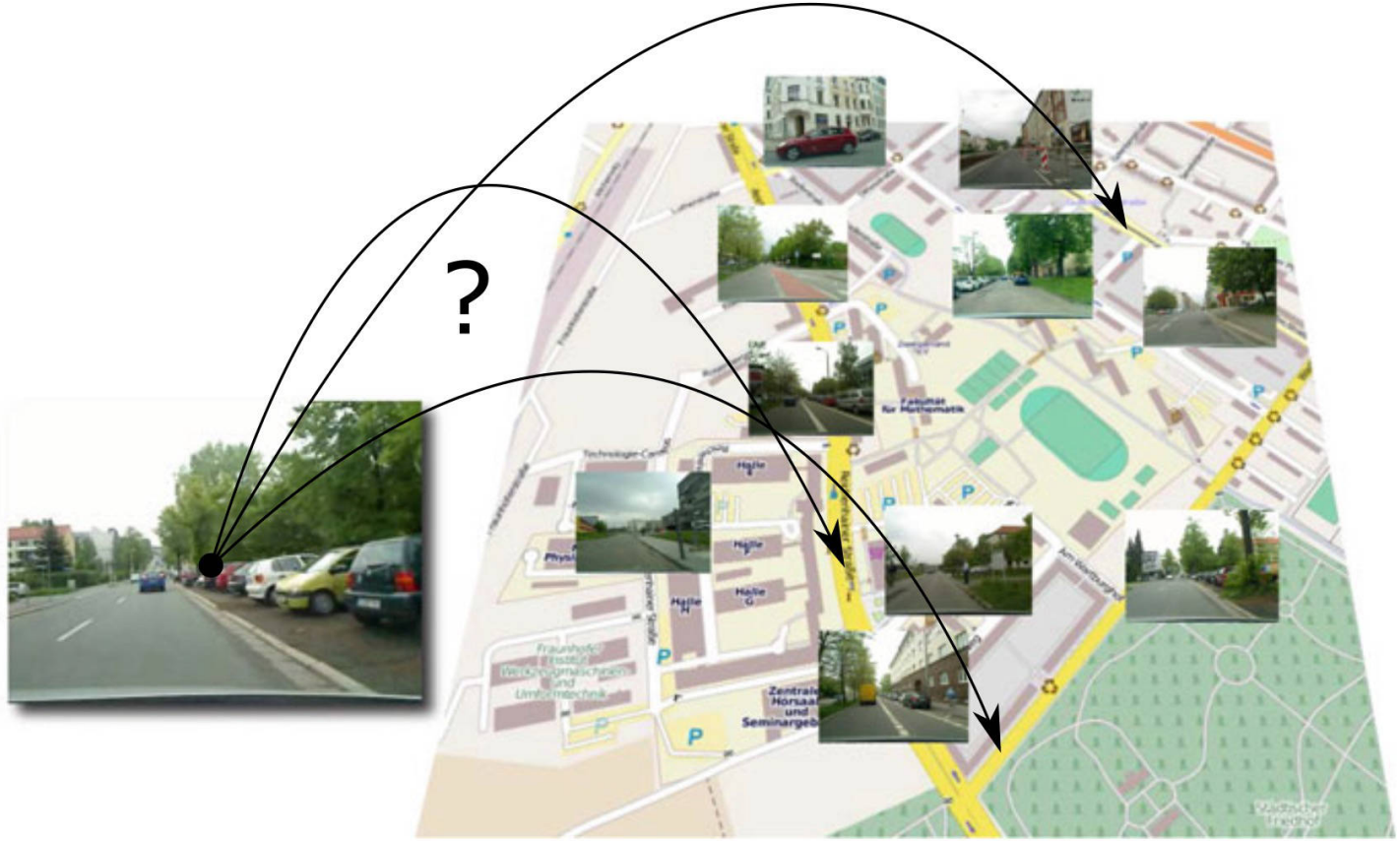
What is place recognition?

Switchable Constraints for Robust Simultaneous Localization and Mapping and Satellite-Based Localization. Niko Suenderhauf, Springer Nature, 2023.

Visual Place Recognition

Given an image of a place, can a robot decide if it has seen this place before?





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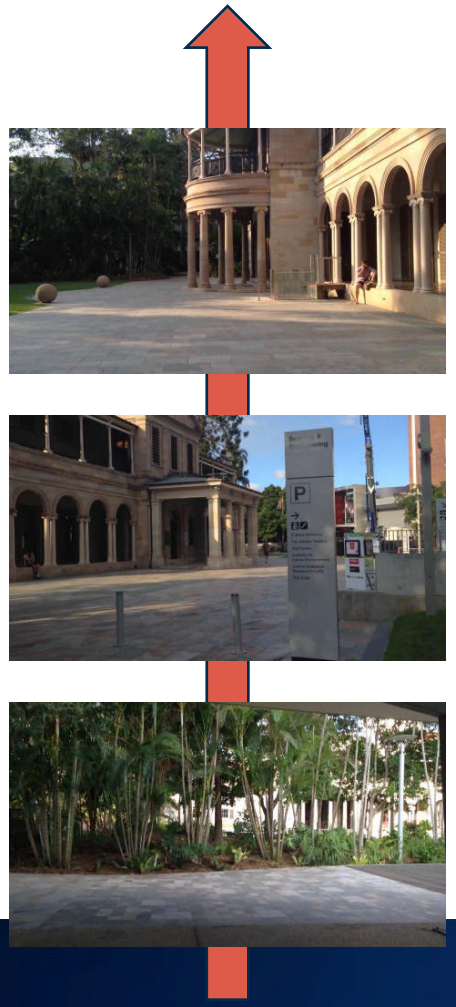
General Overview



- Robot traverses environment for the first time.
- Images (e.g. keyframes) are stored.
- This is the ”reference” / “map” / “database” trajectory.

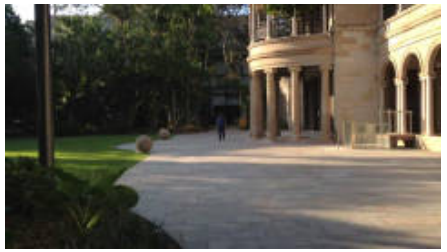
CRICOS No.00213J

General Overview







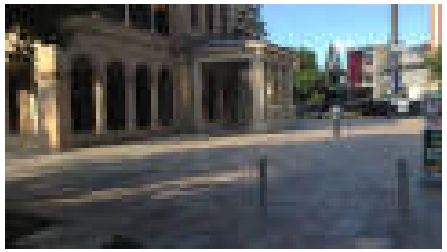
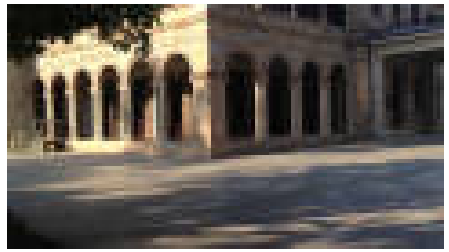
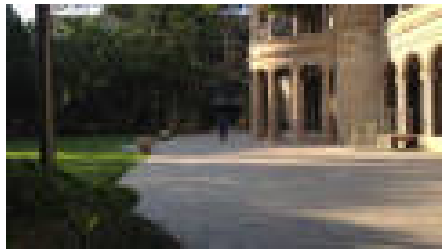


- Later, the robot traverses the environment for the again.
- This is called the “query” trajectory.

- We now want to decide for each new image:
  - Has this place been visited before? OR
  - Is this a new place?

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General Overview

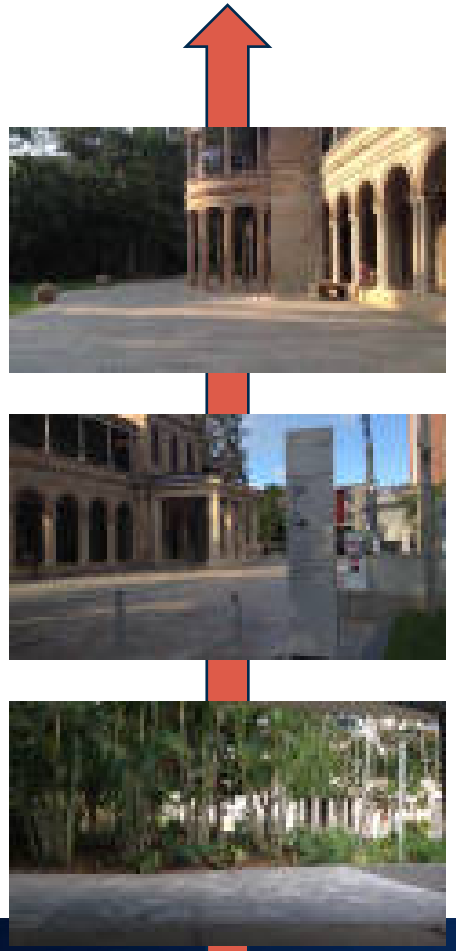






- Notice: The database and query traverses could both be built simultaneously on the go as the robot explores a new environment!

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General Overview









- Idea: Measure pairwise Visual Similarity between a “query” image and the “reference” images.
  - Similarity measure is based on some form of representation or descriptor for the images.

- Q: Why is using the pixel-wise difference as a similarity metric not a good idea?

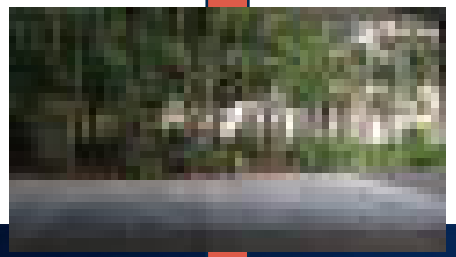
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General Overview









Low Similarity

Low Similarity

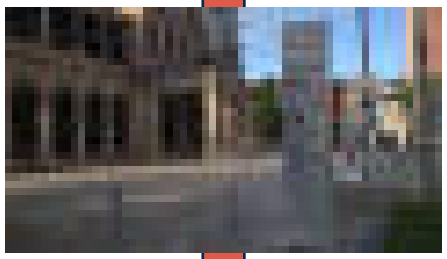
Low Similarity

CRICOS No.00213J

General Overview





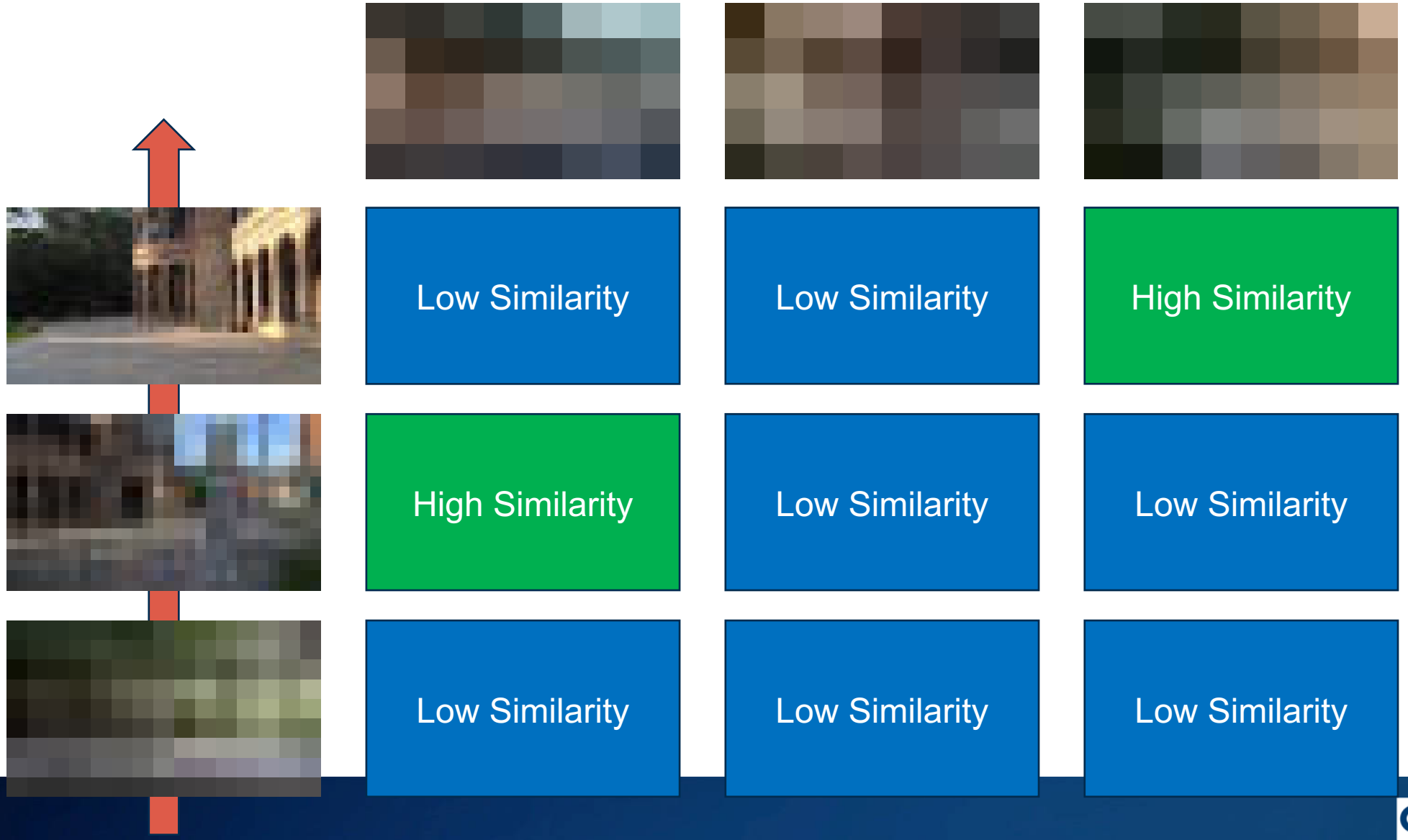


High Similarity | Low Similarity | Low Similarity



Low Similarity | Low Similarity | Low Similarity

General Overview

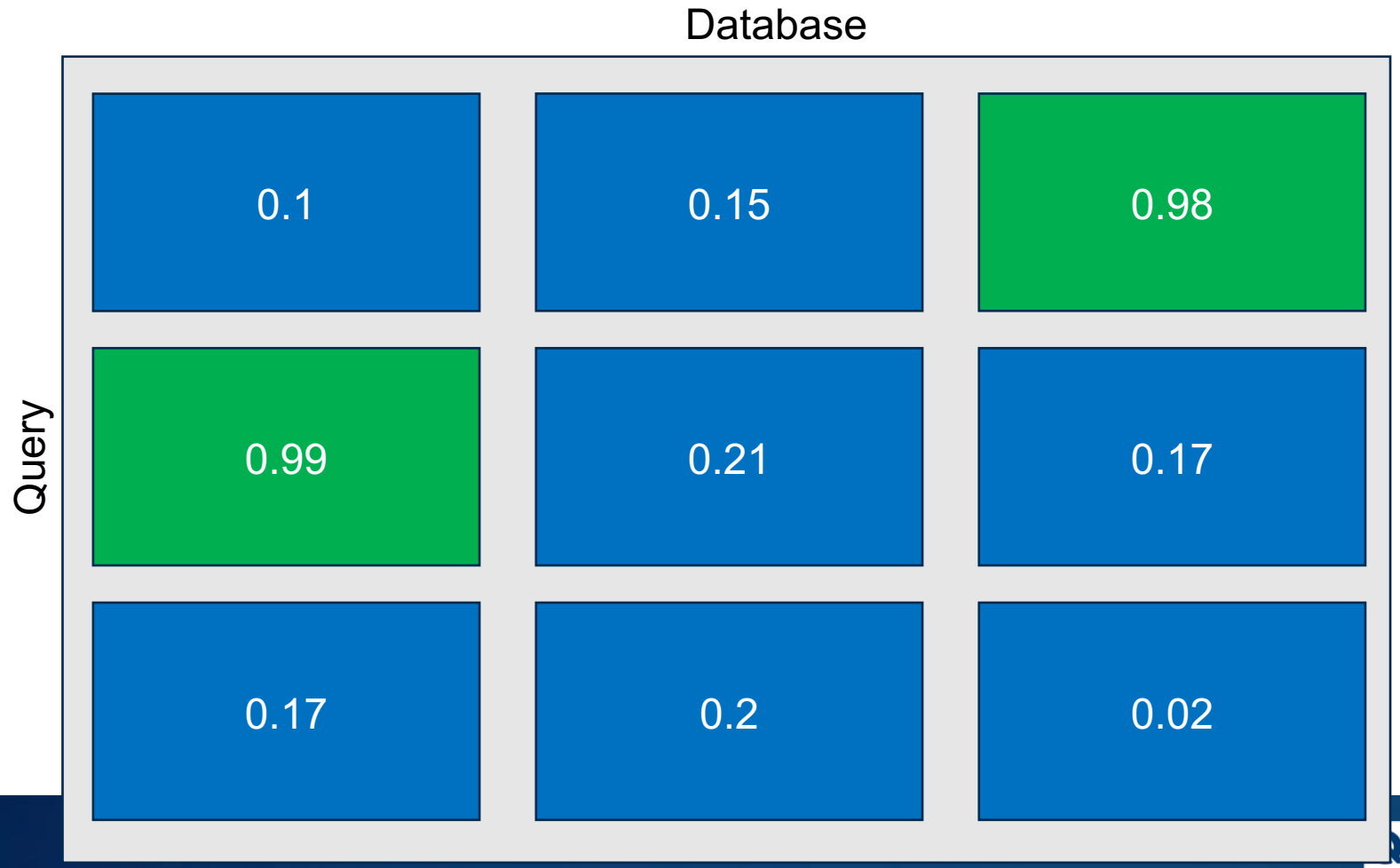


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General Overview

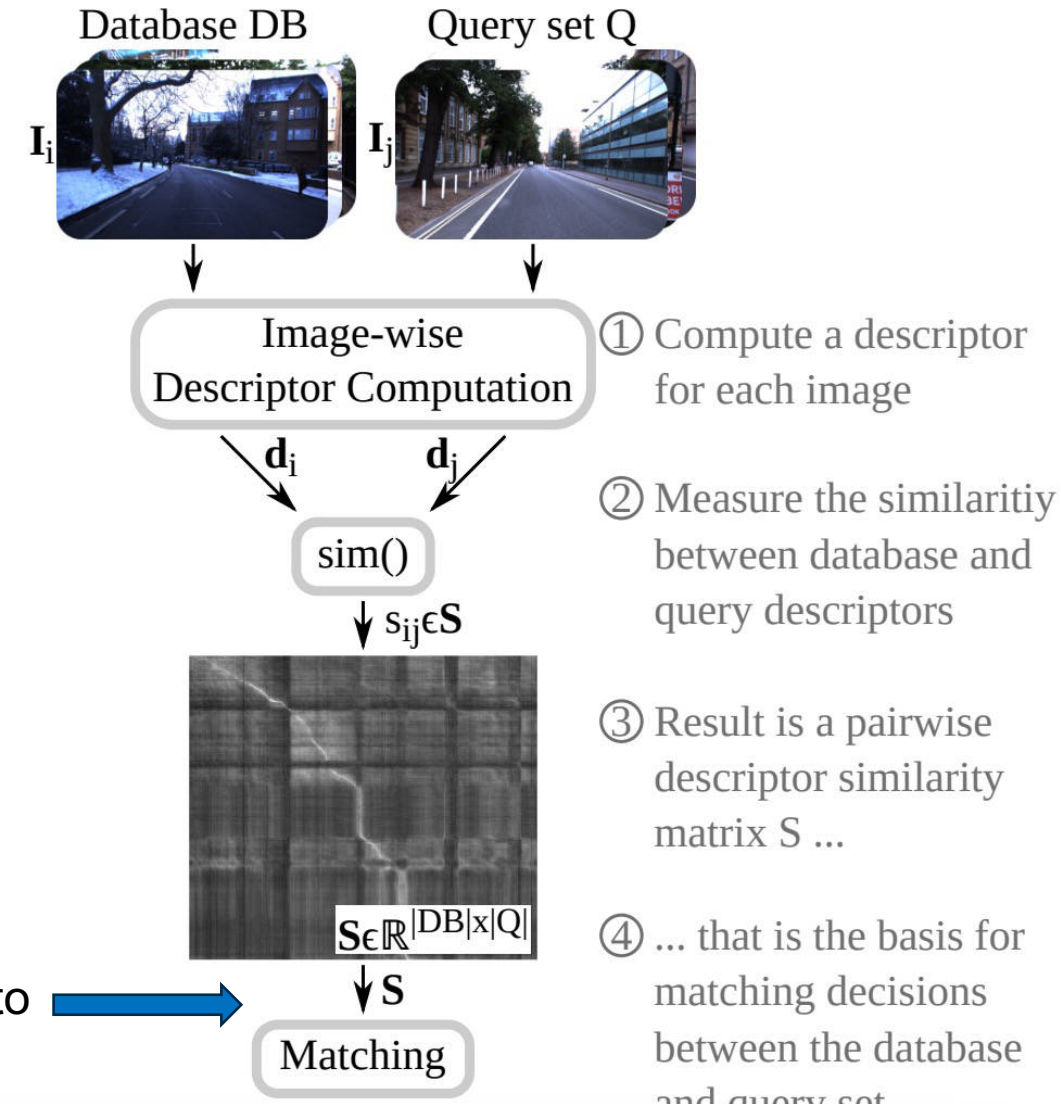
This is called a “similarity matrix”.

A distance matrix is also used, which is the inverse of the similarity.



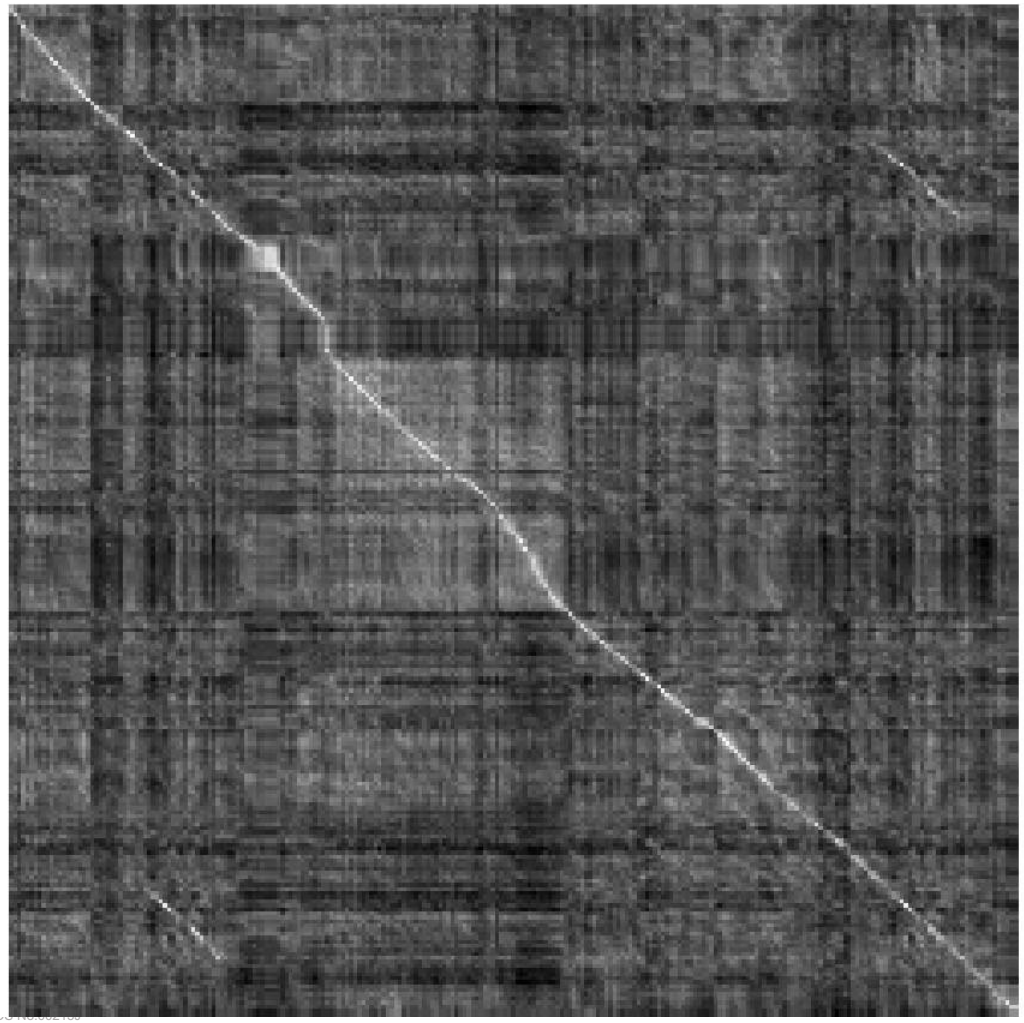
Visual Place Recognition: A Tutorial. Schubert et al.
IEEE Robotics and Automation Magazine. 2023

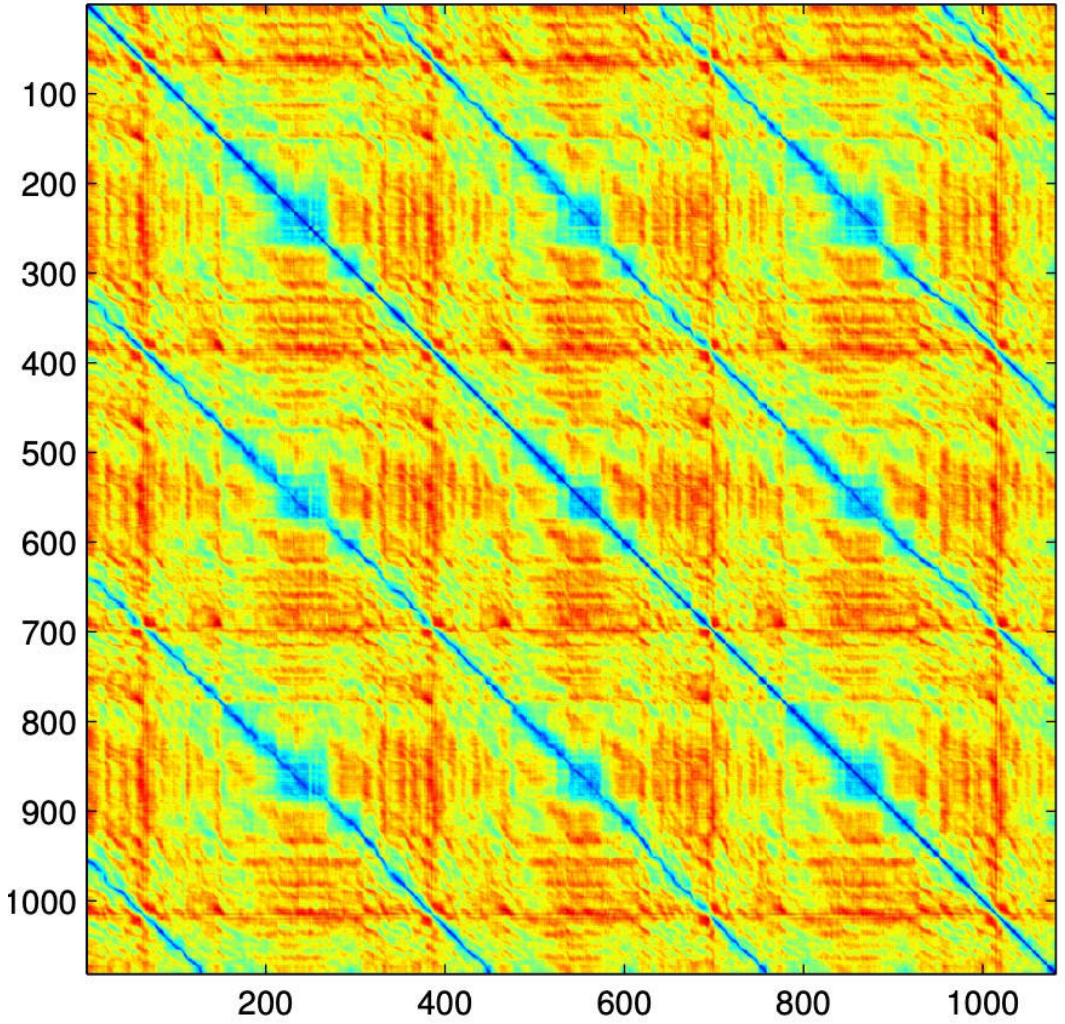
General Overview



Use a threshold, i.e. the similarity has to be bigger than θ to declare a match.

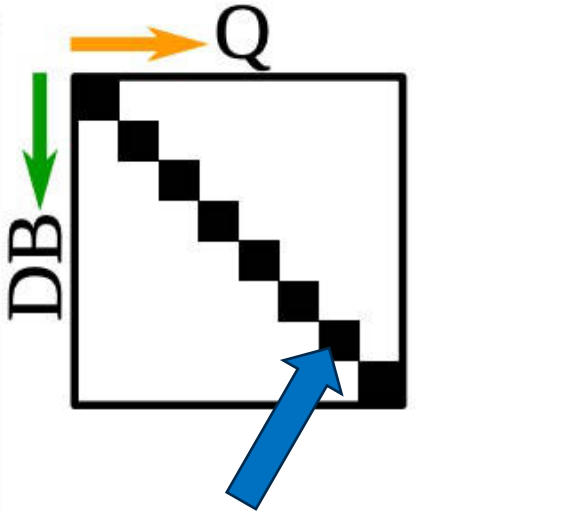
General Overview





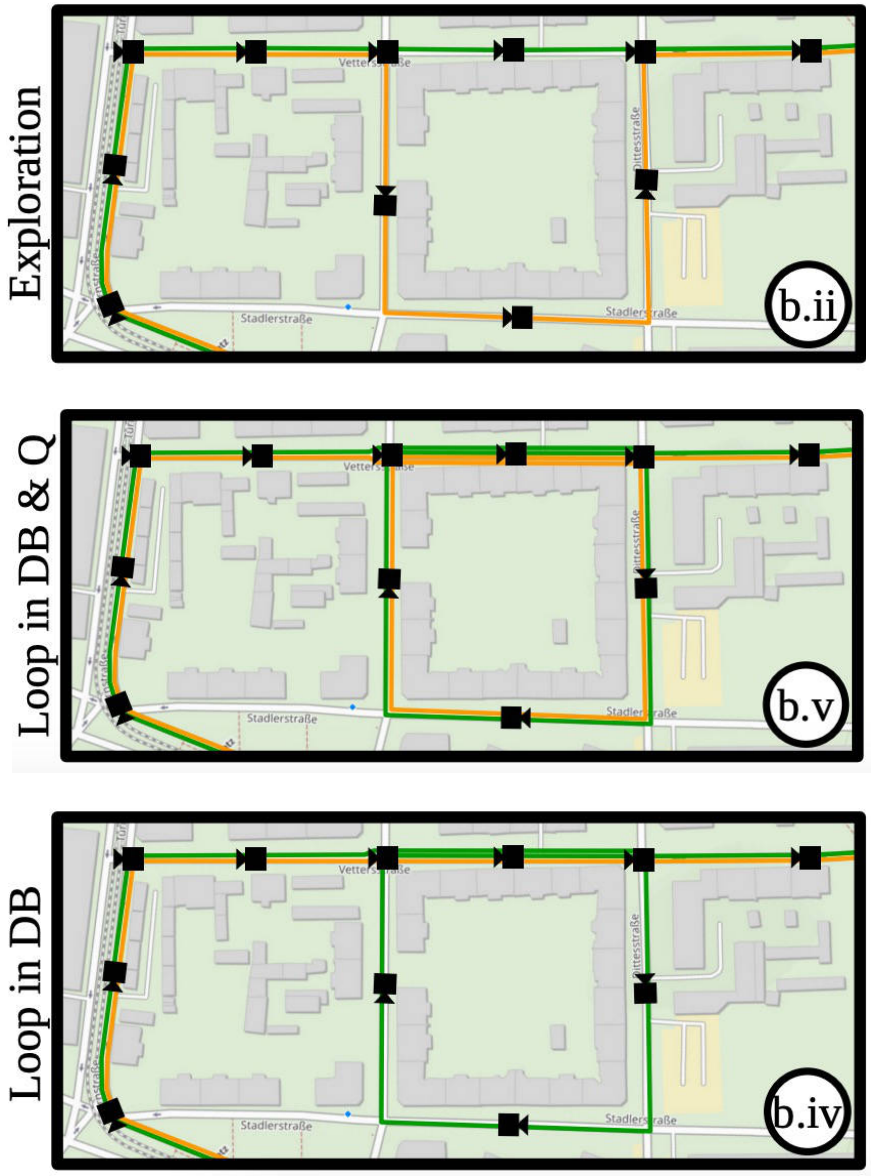
Similarity Matrix Patterns



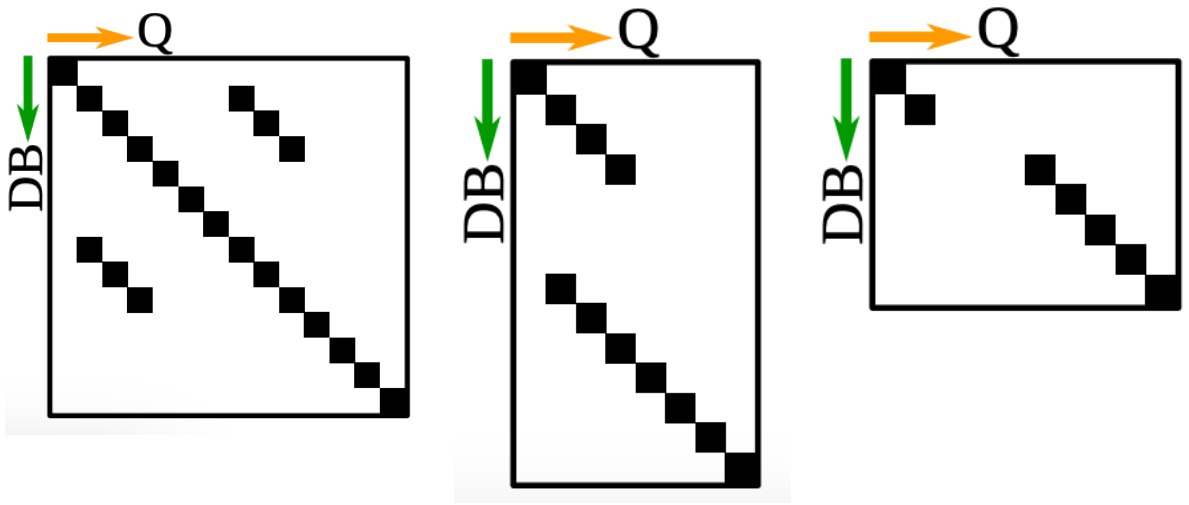


black squares =
ground truth matches

Visual Place Recognition: A Tutorial. Schubert et al.
IEEE Robotics and Automation Magazine. 2023



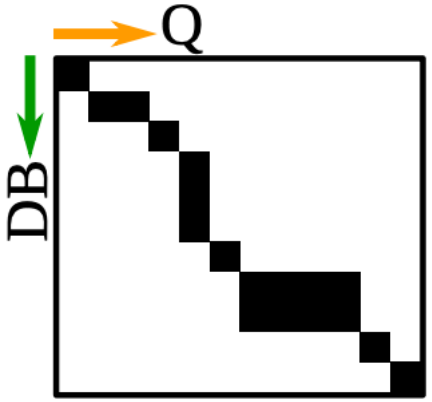
Find the correct similarity matrices for each of the 3 scenarios.



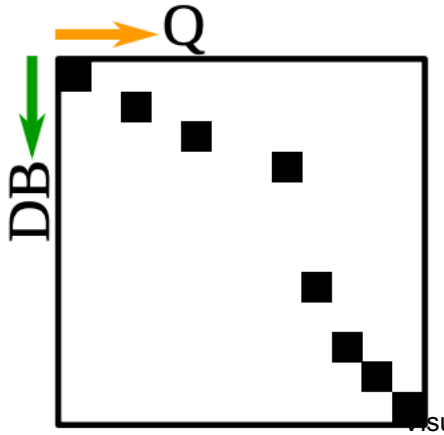
Visual Place Recognition: A Tutorial. Schubert et al. IEEE Robotics and Automation Magazine. 2023

Similarity Matrix Patterns









Visual Place Recognition: A Tutorial. Schubert et al. IEEE Robotics and Automation Magazine. 2023

Single Session VPR Example



Task:

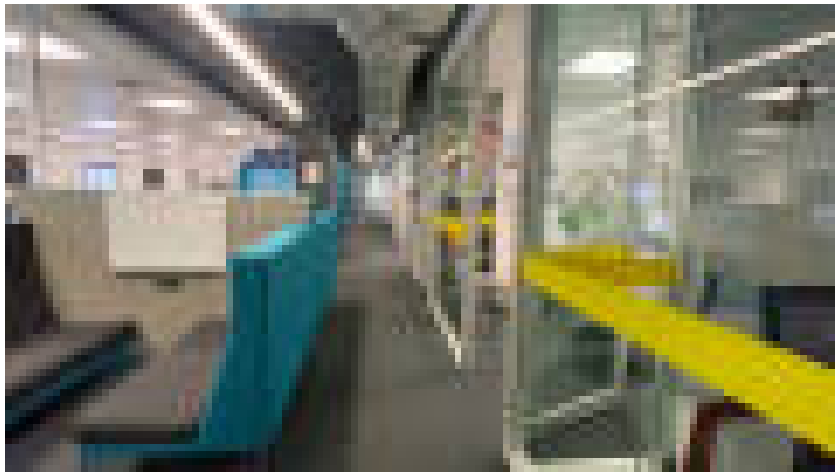
Recognise already-visited places to detect loop closures as the robot explores an environment.

Database and Query are the same, and both growing.

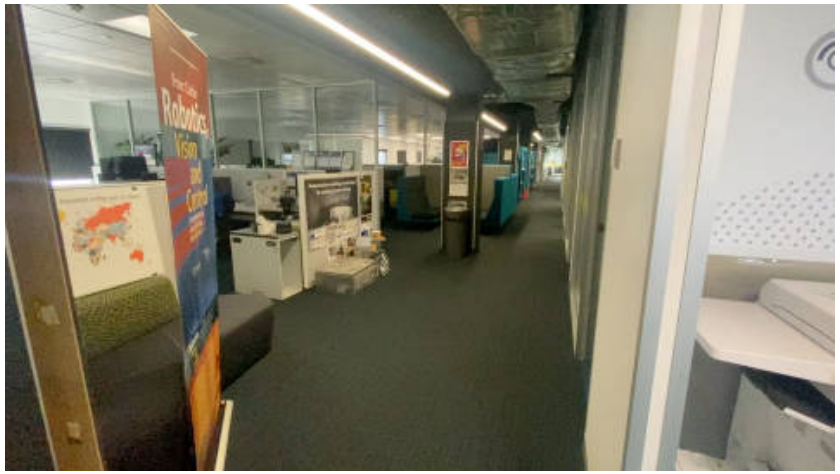
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Multi Session VPR Example

Yesterday



Today



Task:

Localise with respect to yesterday's trajectory.

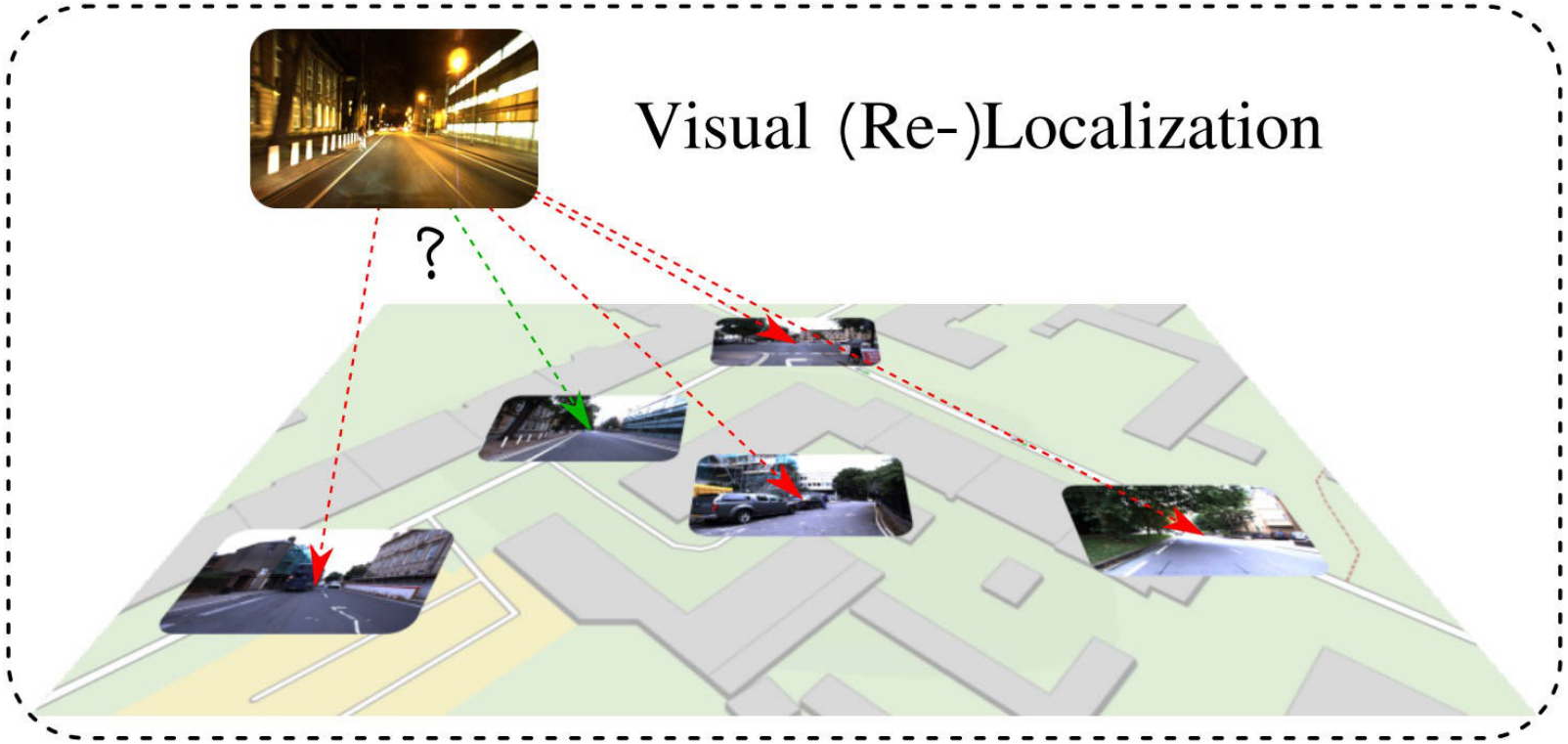
Database: Yesterday, constant.

Query: Today, growing

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Visual Place Recognition: A Tutorial. Schubert et al.
IEEE Robotics and Automation Magazine. 2023

Scenarios

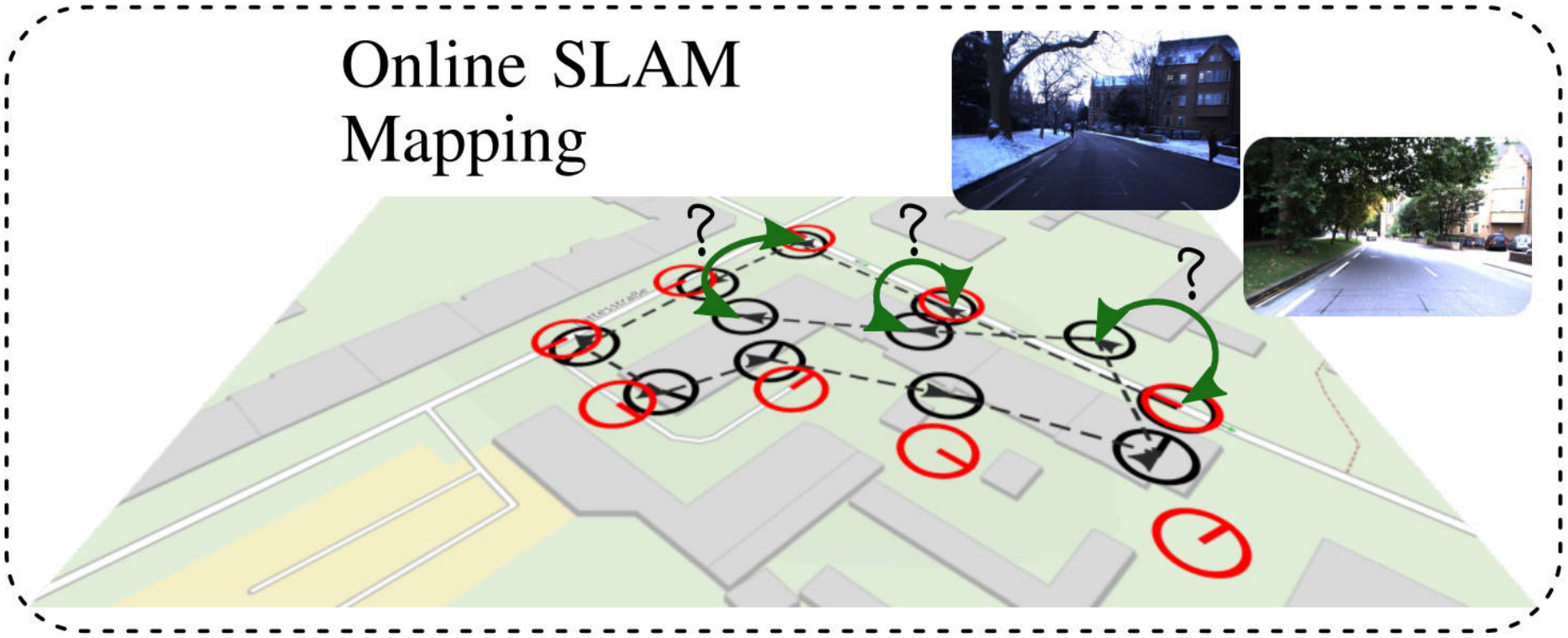


VPR is used to select a small set of candidate images from the database for visual (re-)localization, which are processed with computationally expensive 6-DoF pose estimation methods.

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Visual Place Recognition: A Tutorial. Schubert et al.
IEEE Robotics and Automation Magazine. 2023

Scenarios

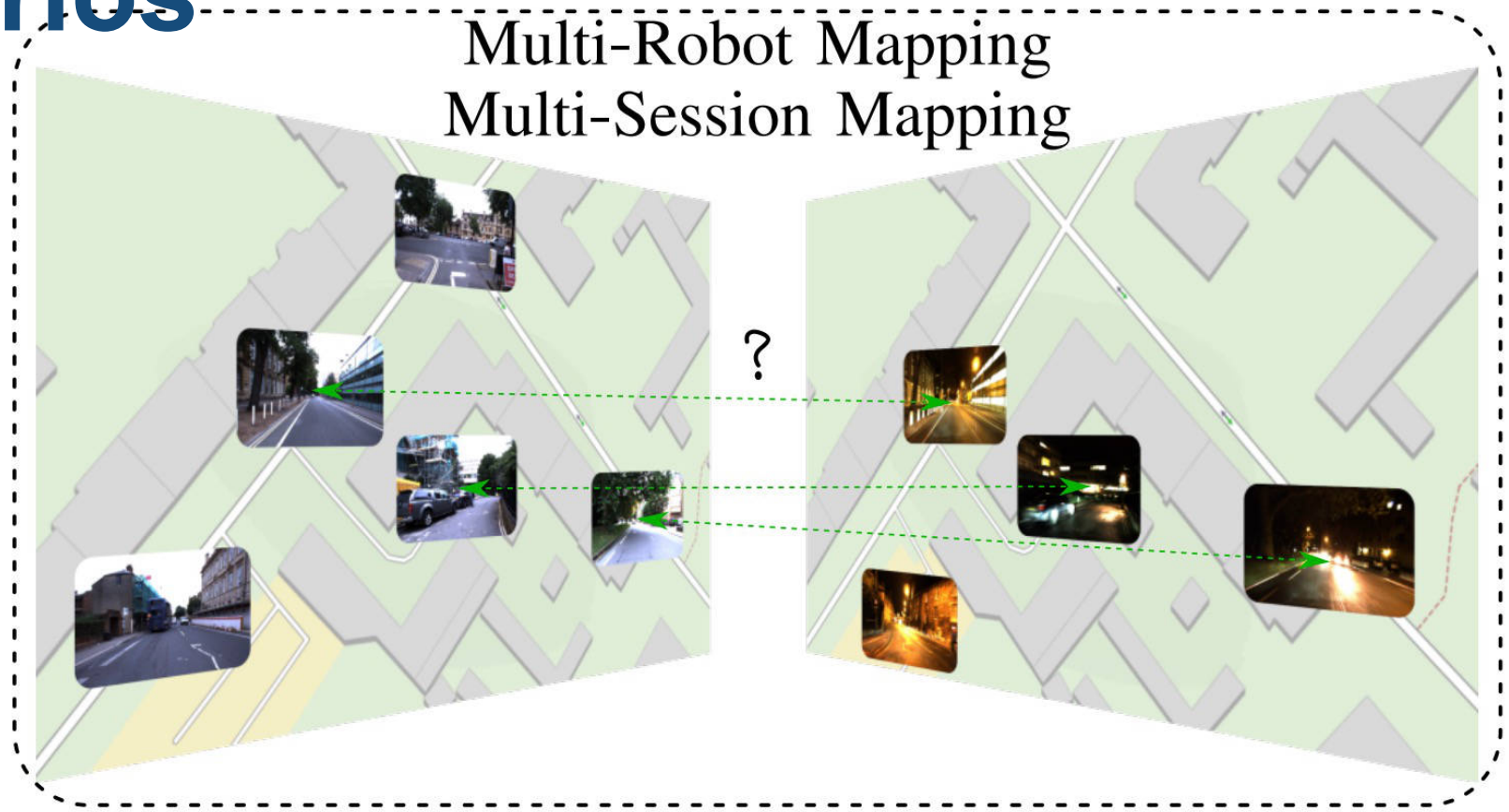


VPR can be used to detect loop closures in a SLAM pipeline (green arrows) or to re-localise after mislocalisation or if the robot was moved (kidnapped robot).

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Visual Place Recognition: A Tutorial. Schubert et al.
IEEE Robotics and Automation Magazine. 2023

Scenarios



In **multisession mapping**, VPR can be used for loop closure detection, but this time in a batch manner where all images are known in advance. In **multi-robot mapping**, VPR is used to merge maps by detecting places that have been visited by multiple robots.

Visual Place Recognition: A Tutorial. Schubert et al.
IEEE Robotics and Automation Magazine. 2023

VPR Taxonomy and Applications

	Online VPR		Batch VPR
Single-Session VPR	Online SLAM		Mapping
Multi-Session VPR	**Database Grows**	**Database constant**	Multi-Session Mapping
	Multi-Robot Mapping	Visual Localisation	

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What makes visual place recognition easy or hard?
Stefan Schubert and Peer Neubert. arXiv:2106.12671, 2021.

VPR is Challenging





Challenges: Viewpoint





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Place Recognition with ConvNet Landmarks: Viewpoint-Robust, Condition-Robust, Training-Free.
Niko Sünderhauf, Sareh Shirazi, Adam Jacobson, Feras Dayoub, Edward Pepperell, Ben Upcroft, Michael Milford. RSS 2015.

Challenges: Dynamic Scenes, Occlusion





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Challenges: Illumination, Weather





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Milford, M. J., & Wyeth, G. F. (2012).
SeqSLAM: Visual route-based navigation for sunny summer days and stormy winter nights.
International Conference on Robotics and Automation (ICRA)

Challenges: Seasons



CRICOS No.00213J

Sünderhauf, N., Neubert, P., & Protzel, P. (2013).
Are we there yet? Challenging SeqSLAM on a 3000 km journey across all four seasons.
In *Proc. of Workshop on long-term autonomy, International Conference on Robotics and Automation (ICRA)*

Visual Place Recognition: A Survey.
Lowry et al., IEEE Transactions on Robotics, 2016.

Challenges: Perceptual Aliasing

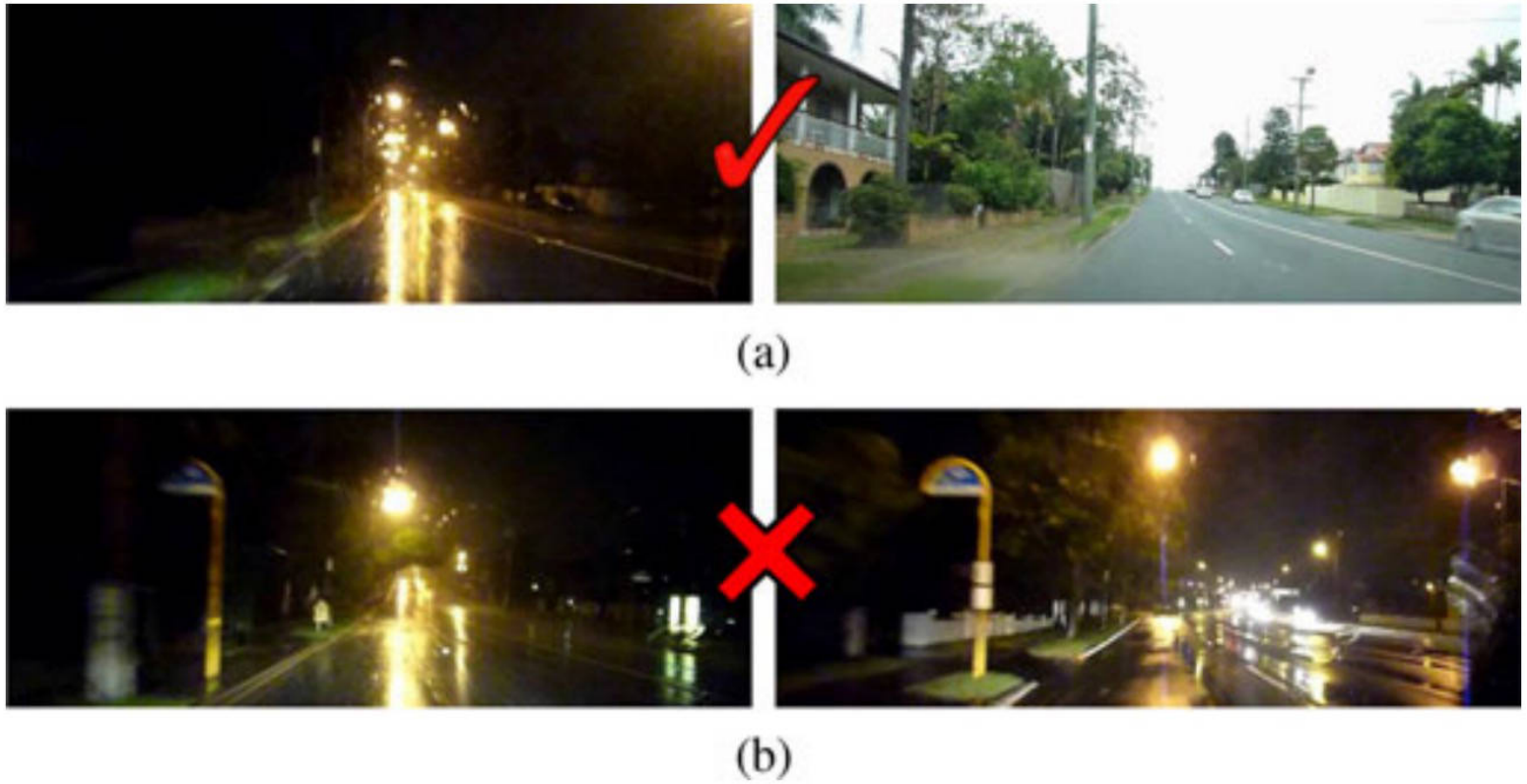
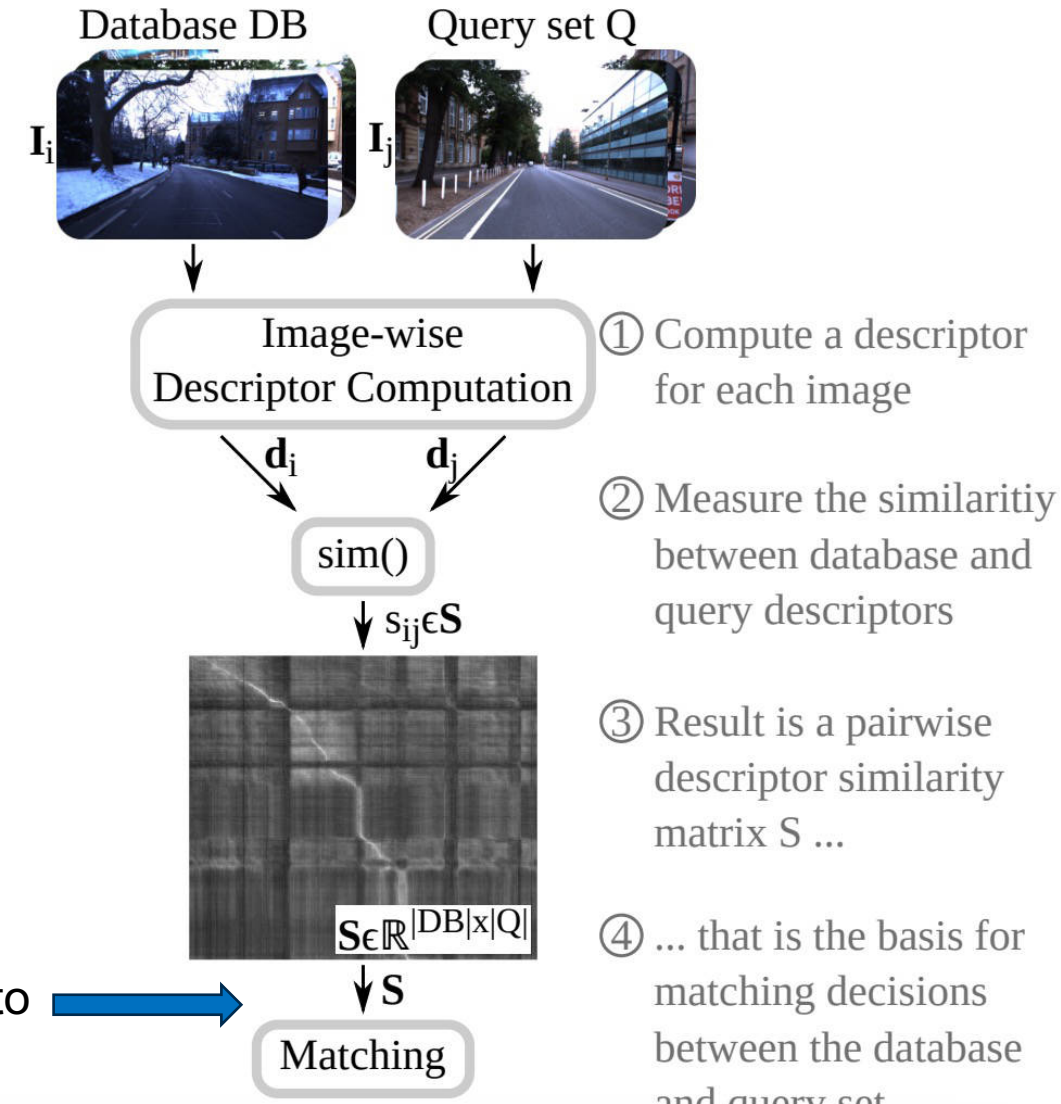


Fig. 1. Visual place recognition systems must be able to (a) successfully match very perceptually different images while (b) also rejecting incorrect matches between aliased image pairs of different places.

Visual Place Recognition: A Tutorial. Schubert et al.
IEEE Robotics and Automation Magazine. 2023

General Overview



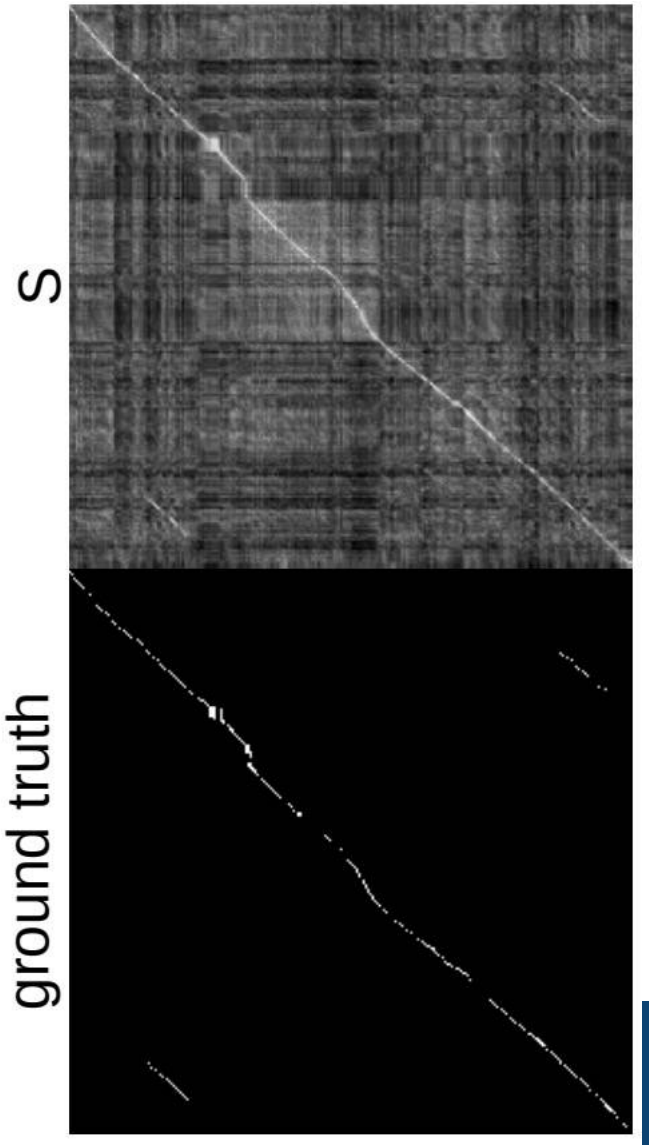
Use a threshold, i.e. the similarity has to be bigger than θ to declare a match.

How do we know it’s good?
VPR Evaluation

Evaluation of Place Recognition Methods

- We need the ground truth matches.
- Then we have a binary (yes/no) classification problem.

- Typical metrics are
  - True Positives (TP)
  - False Positives (FP)
  - False Negatives (FN)
  - Precision (P)
  - Recall (R)

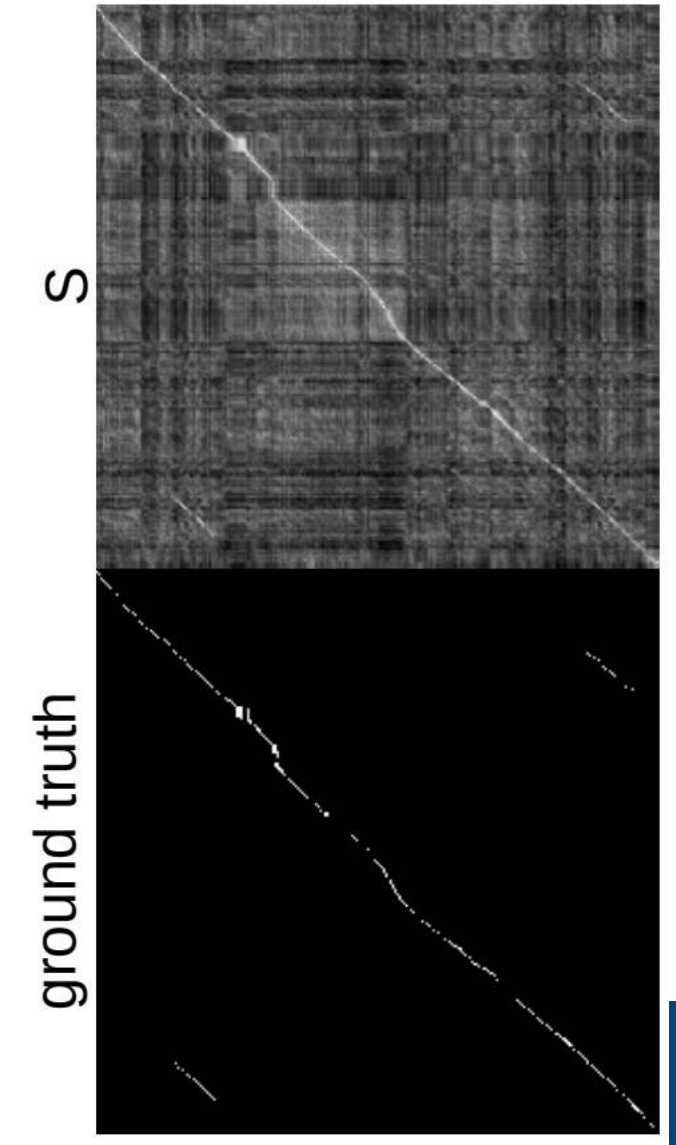


Evaluation of Place Recognition Methods

- **Precision:**
  - The ratio of correctly matched images to the total number of matched images.

$$P = \frac{\#TP}{\#TP + \#FP}$$

  - Example:
    - Your algorithm finds 100 matches, but only 65 are actually correct.
    - How many True Positives and False Positives?
    - What is the Precision?



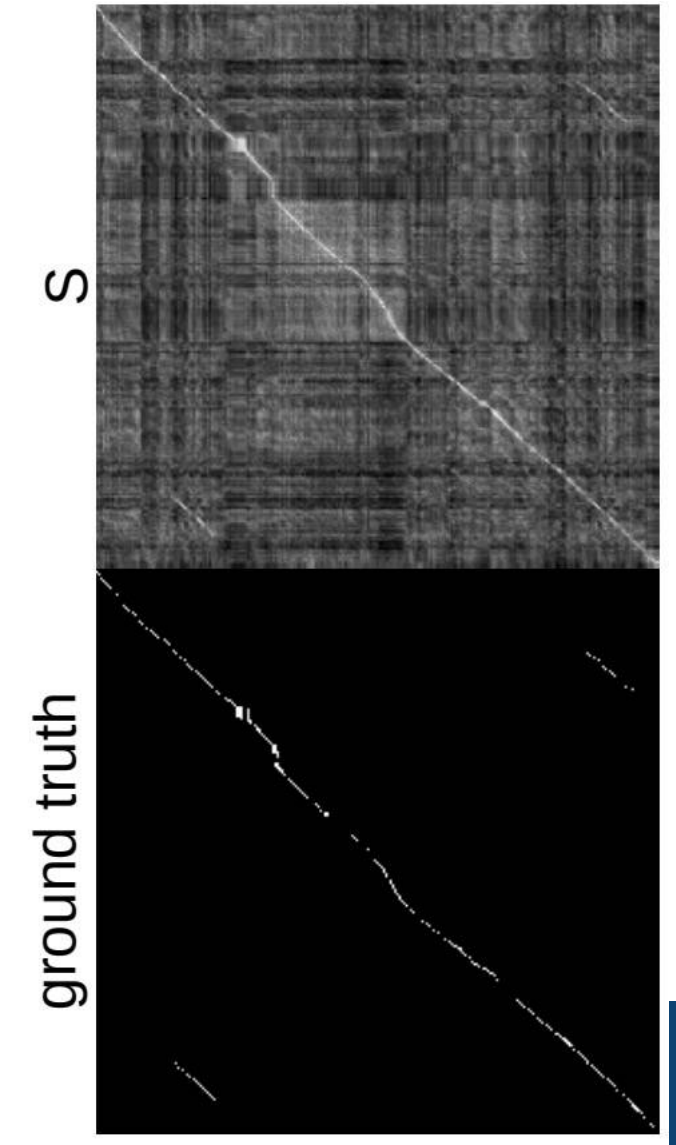
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Evaluation of Place Recognition Methods

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$$P = \frac{\#TP}{\#TP + \#FP}$$

  - Example:
    - Your algorithm finds 100 matches, but only 65 are actually correct.
    - How many True Positives and False Positives?
    - What is the Precision?
    - TP = 65 FP = 35 P = 0.65

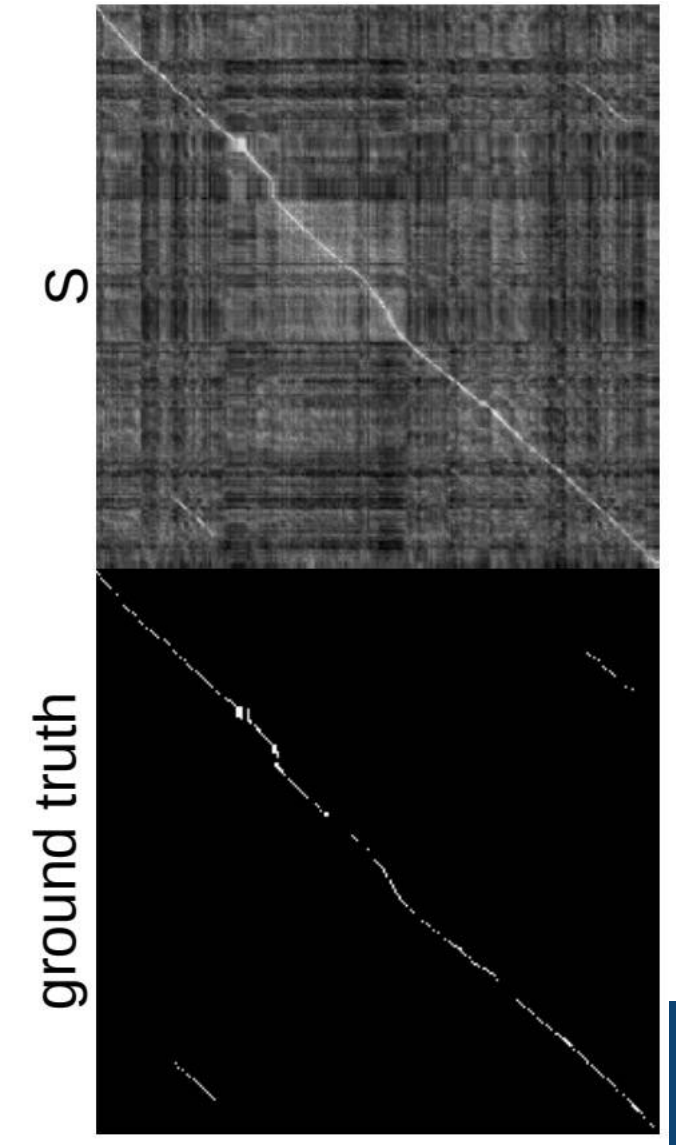


Evaluation of Place Recognition Methods

- **Recall:**
  - The ratio of correctly matched images to the total number of ground truth positives (GTP).

$$R = \frac{\#TP}{\#GTP}$$

  - GTP is number of query images for which a ground truth match exists. (more nuanced: see tutorial paper)
  - Example:
    - Your algorithm finds 100 matches, but 200 images had ground truth matches.
    - What is R?



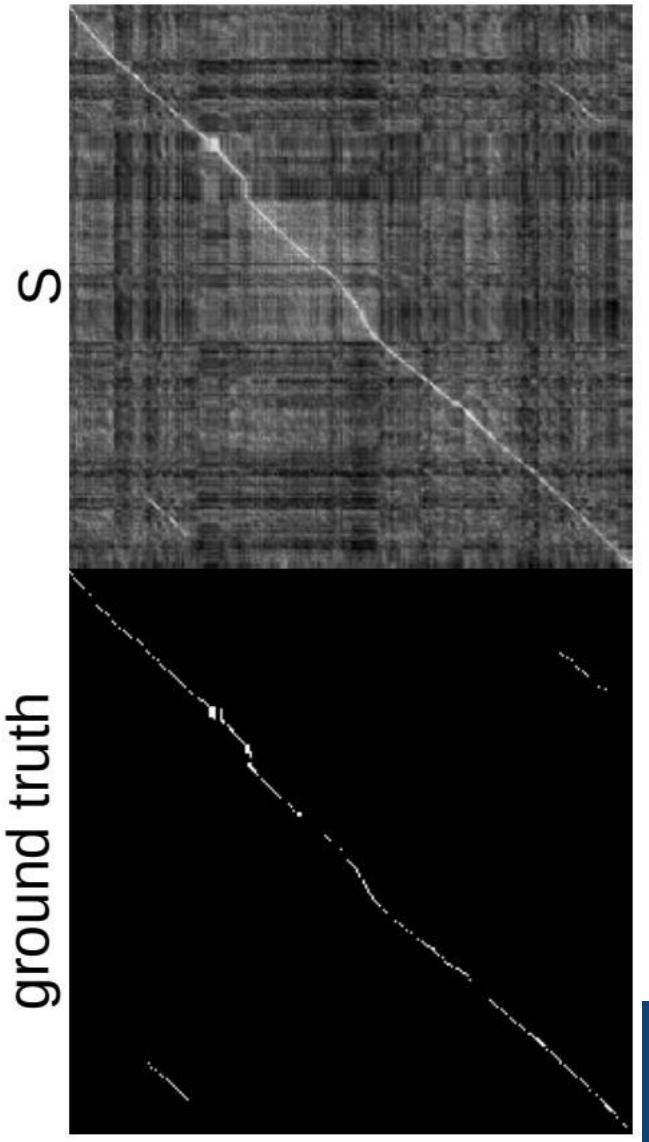
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Evaluation of Place Recognition Methods

- **Recall:**
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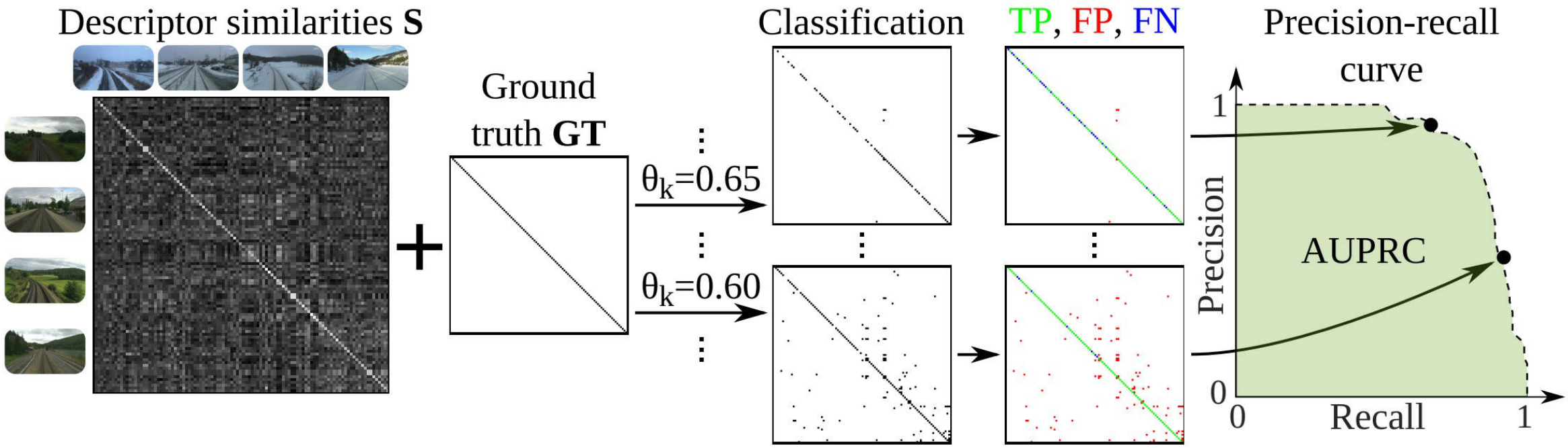
$$R = \frac{\#TP}{\#GTP}$$

  - GTP is number of query images for which a ground truth match exists. (more nuanced: see tutorial paper)
  - Example:
    - Your algorithm finds 100 matches, but 200 images had ground truth matches.
    - What is R?
    - R = 100/200 = 0.5



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Evaluation Pipeline



Other evaluation metrics:

- Maximum recall at 100%, 99%, 95% Precision
- AUPRC = Area under Precision-Recall-Curve
- Maximum F1 score (harmonic mean of precision and recall = $2\frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$

Visual Place Recognition: A Tutorial. Schubert et al. IEEE Robotics and Automation Magazine. 2023

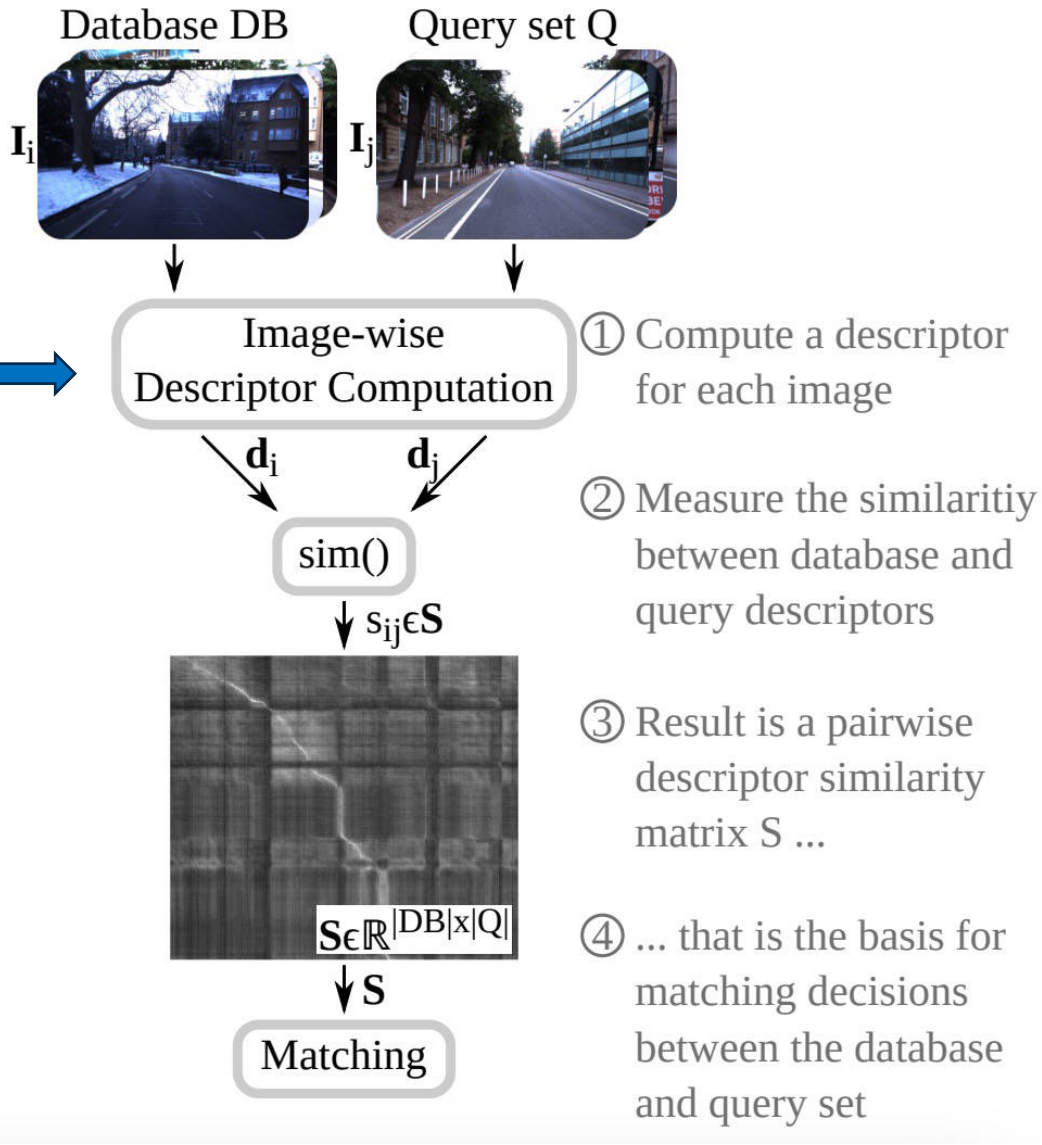
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How do we actually do it?
VPR Methods

Visual Place Recognition: A Tutorial. Schubert et al.
IEEE Robotics and Automation Magazine. 2023

Descriptors for VPR

How do we do this?



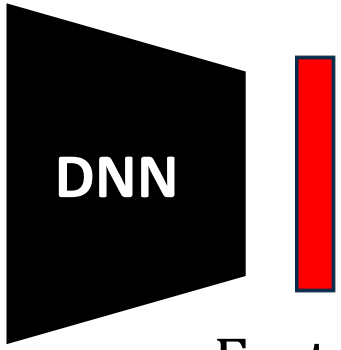
Two Ideas:

- **Holistic / Global Descriptor**
  - One descriptor for the whole image
- **Local Descriptors**
  - A set of descriptors for an image

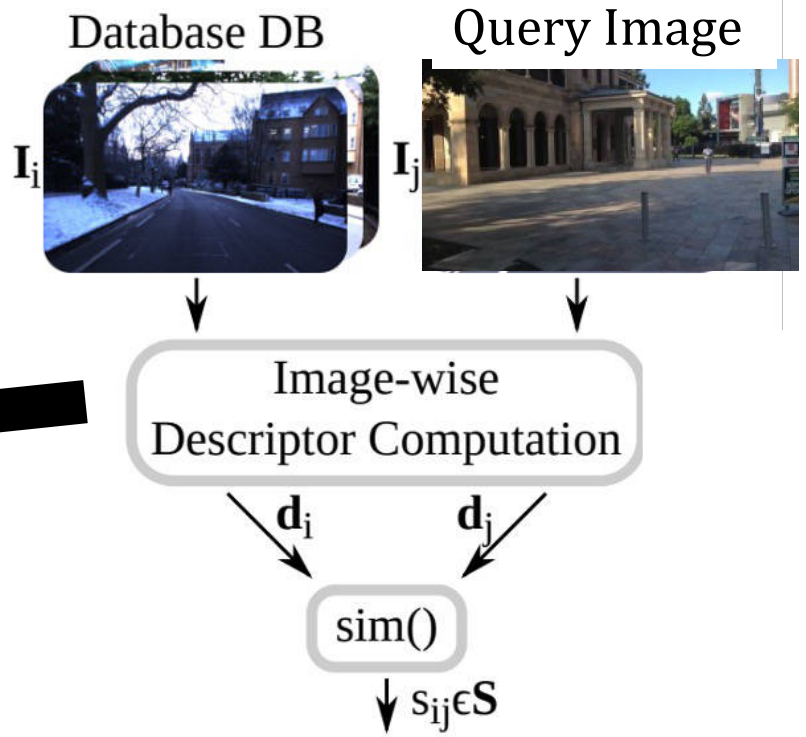
General VPR Pipeline with Image Retrieval

Global image descriptor





Feature descriptor





Most similar image in database

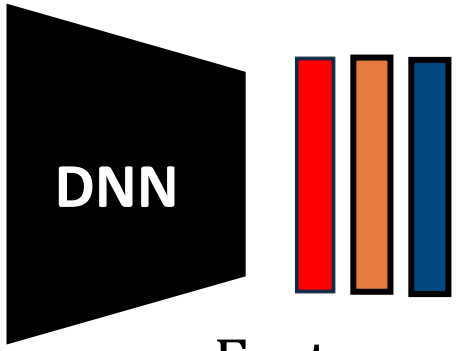
Advantage:

- compact representation
- fast retrieval

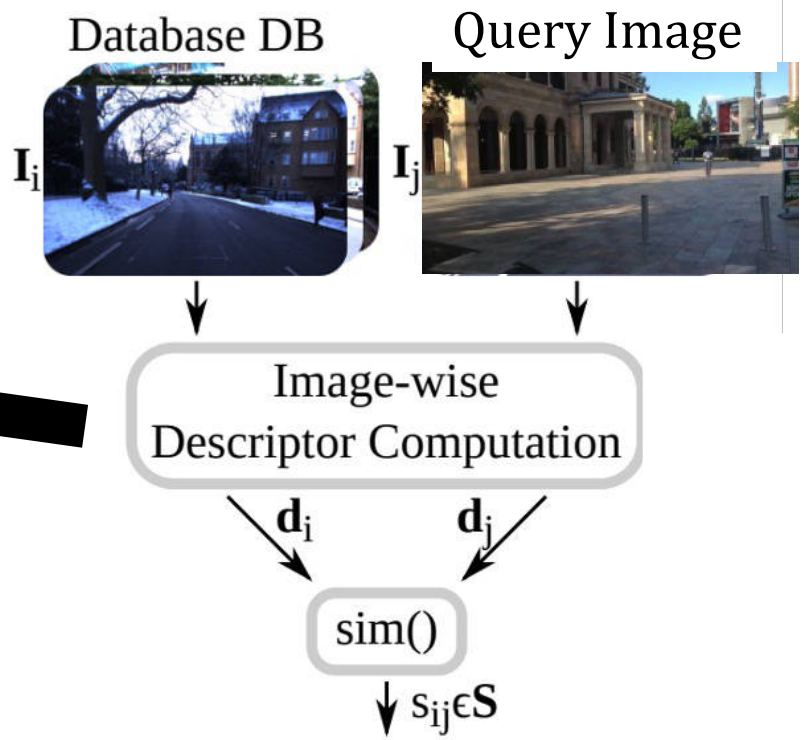
General VPR Pipeline with Image Retrieval

Local image descriptors





Feature descriptors





Most similar image in database

Advantage:

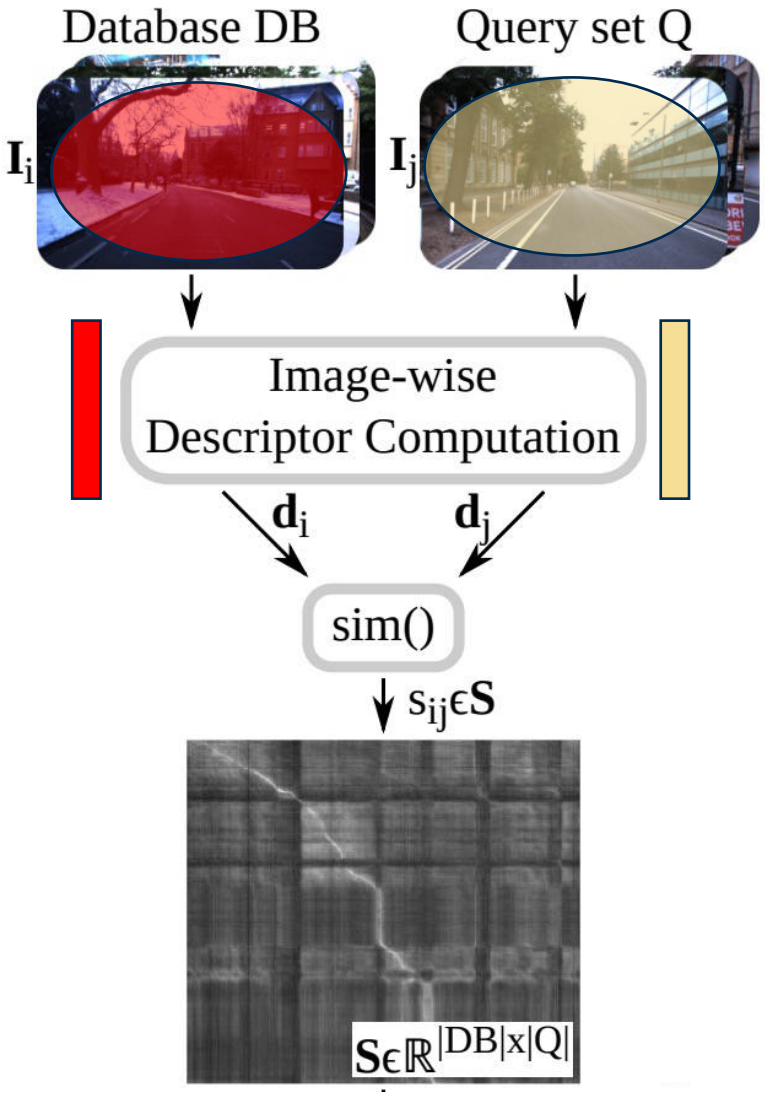
- preserves spatial/local detail
- can handle viewpoint changes better
- but matching is more expensive

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Global Descriptors

- Extract a **single global** descriptor for the image.
- Compare the descriptor vectors.

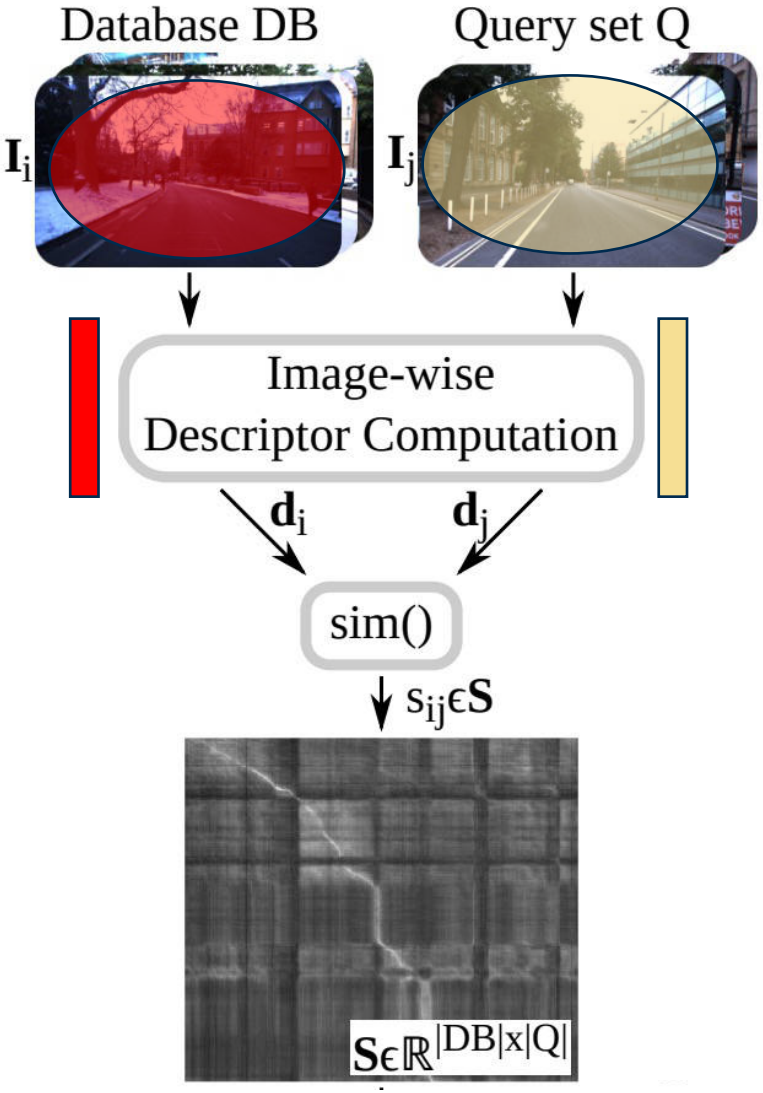
- Before deep learning, hand-crafted global descriptors were used:
  - A. Oliva and A. Torralba. Building the gist of a scene: The role of global image features in recognition. Visual Perception, Progress in Brain Research, 155, 2006.
  - BRIEF-Gist (Suenderhauf & Protzel, IROS 2011)
    - Simple idea: Calculate a BRIEF descriptor for every image.

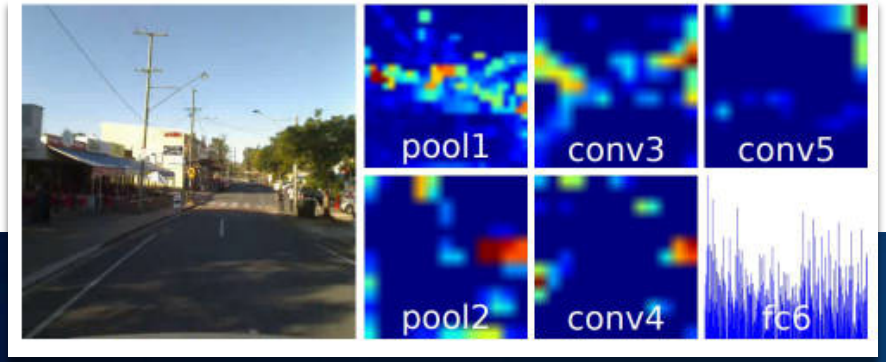


Global Descriptors

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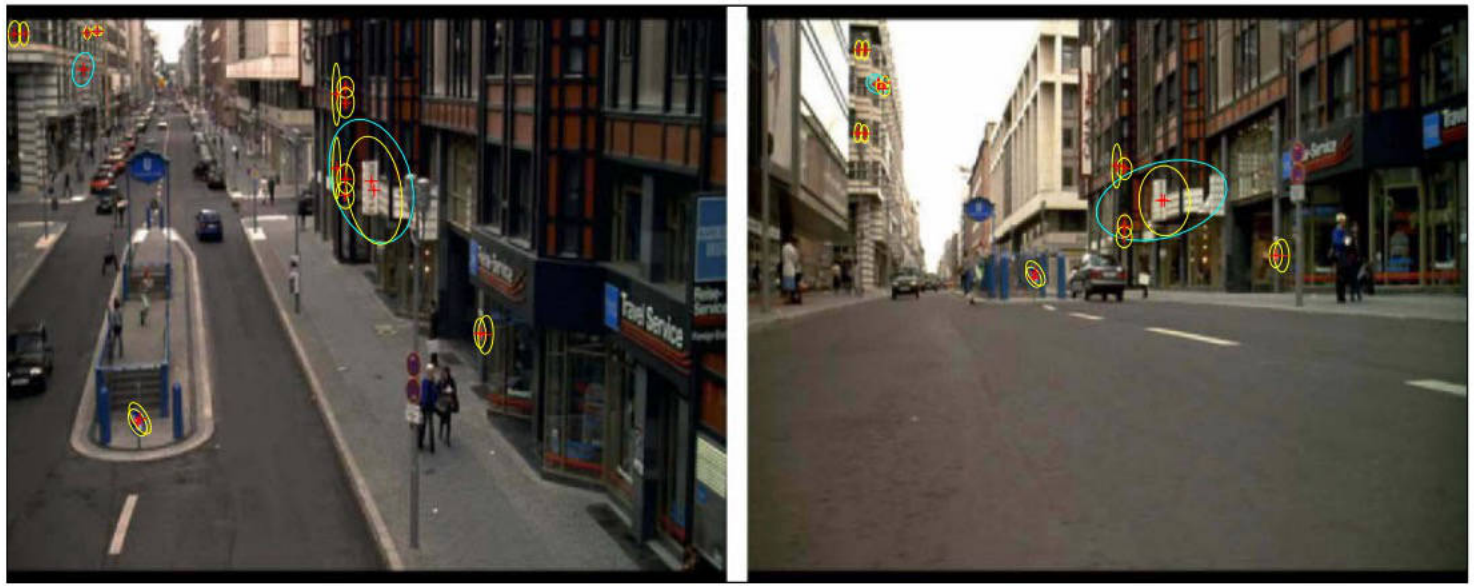
- After deep learning: use network features
  - On the Performance of ConvNet Features for Place Recognition (Suenderhauf, Shirazi, Dayoub, Upcroft, Milford. IROS 2015.)
    - Simple idea: Take a feature map from a CNN and use it as a global descriptor for every image.



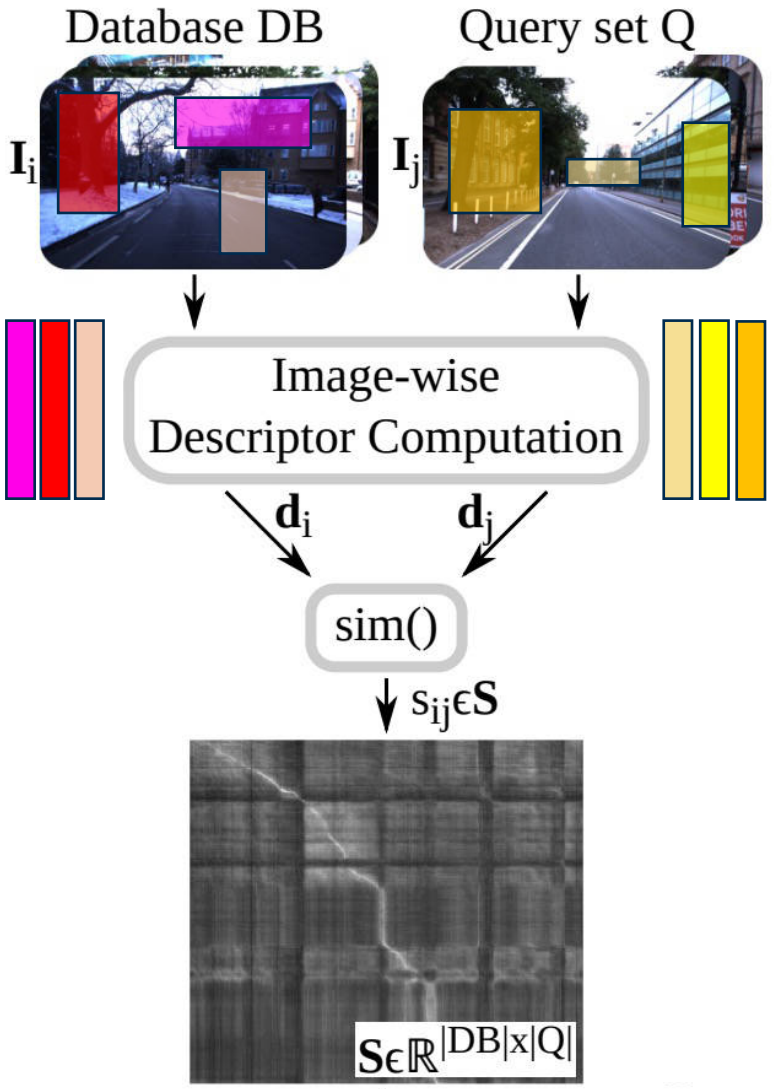


Local Descriptors

- Extract **multiple local** descriptors in the image.
- Compare sets of descriptor vectors.



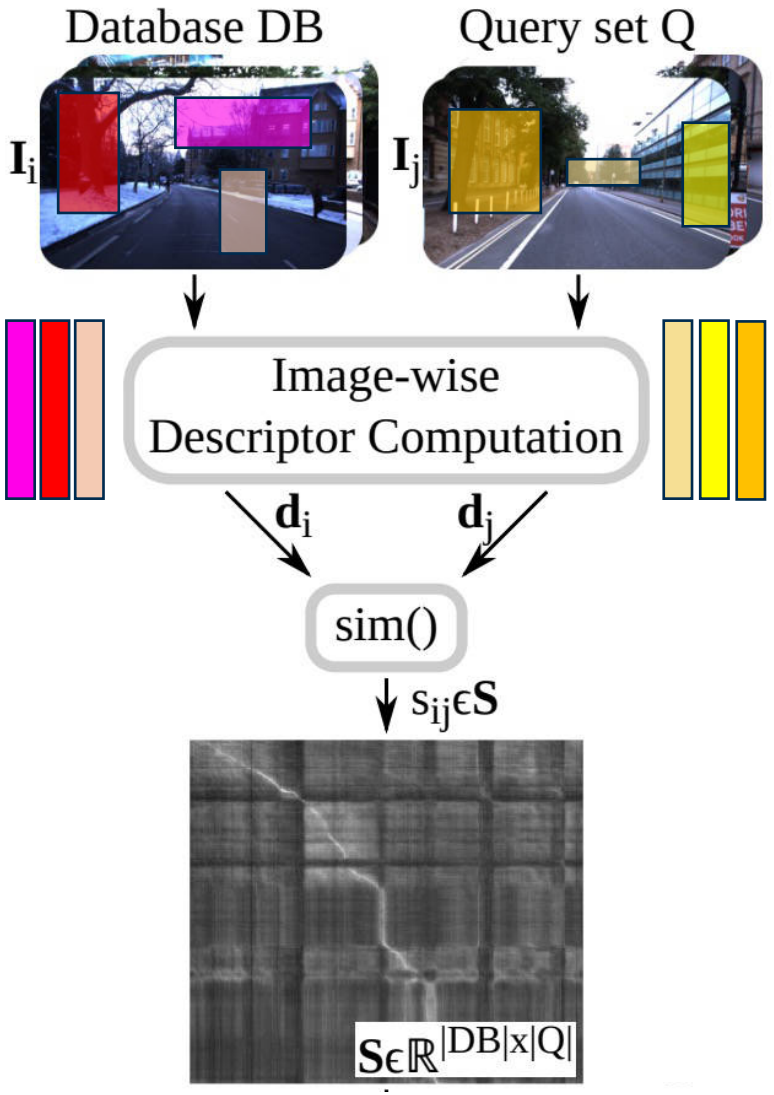
Video Google: A Text Retrieval Approach to Object Matching in Videos. Josef Sivic and Andrew Zisserman. ICCV 2003.



Local Descriptors

- Extract **multiple local** descriptors in the image.
- Compare sets of descriptor vectors.

- We already know how to describe local image regions! → ORB, SIFT …
- But: How do we turn a variable-sized set of local descriptors into something that can be efficiently retrieved?

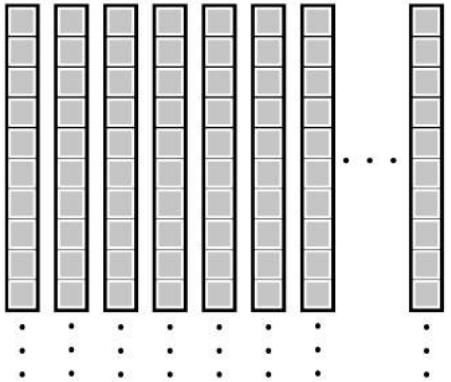


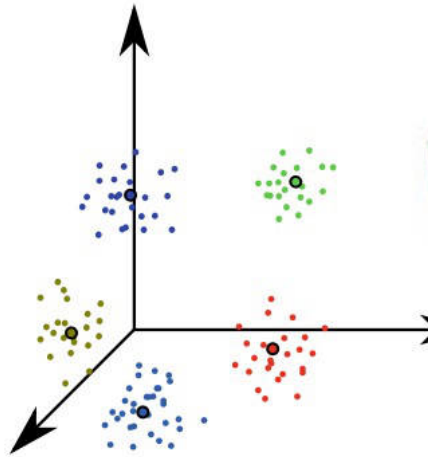
Local Descriptors: Bag of Visual Words



1. Collect Keypoints from Training Set

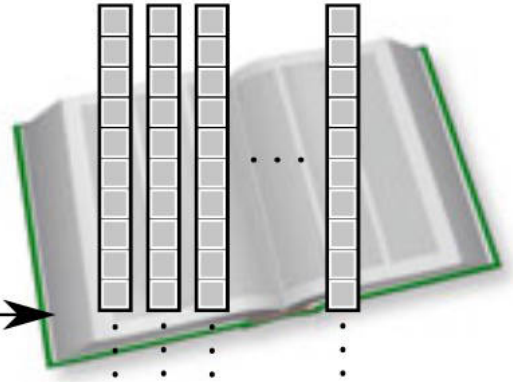
2. Calculate Keypoint Descriptors





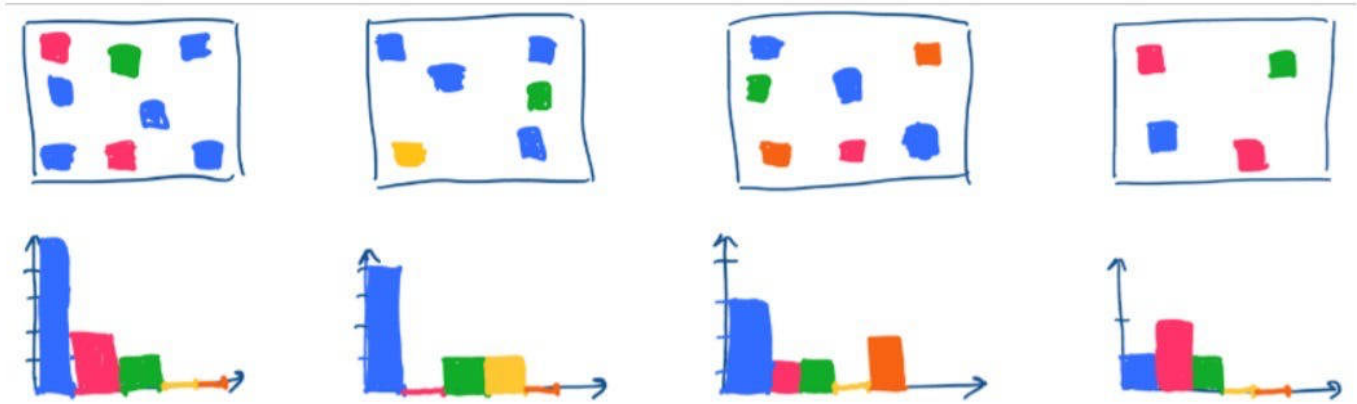
3. Cluster in Descriptor Space

4. Keep Cluster Centers as Visual Words



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Local Descriptors: Bag of Visual Words



[Image Courtesy: Olga Vysotska]

During deployment: extract features, find closest visual word for each of them, then compare two images based on appearance of words.

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Visual Words

Just like for text documents, some words are more informative than others.
E.g. some words are very common in the environment.





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Term Frequency – Inverse Document Frequency

How often does word i appear in document d?

How many documents are there?

$$t_i = \frac{n_{id}}{n_d} \log \frac{N}{n_i}$$

How many words are in document d?

How often does word i appear across all documents?

Term frequency

Inverse document frequency

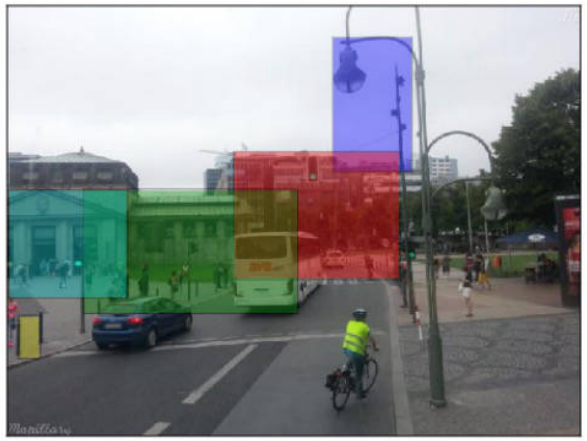
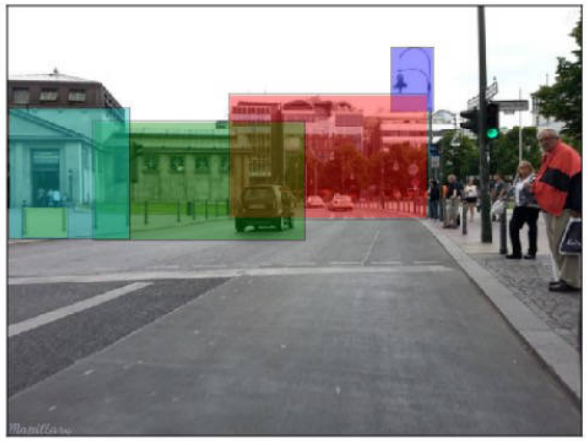
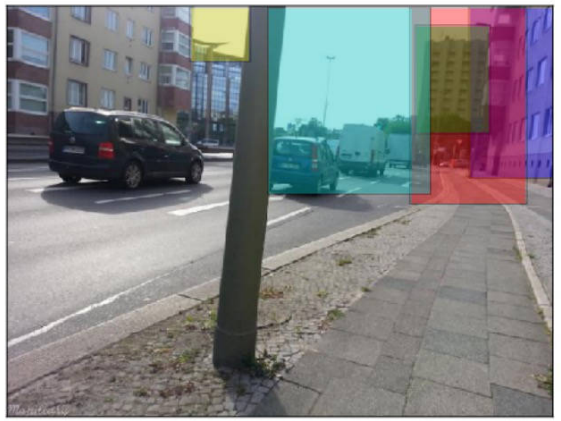
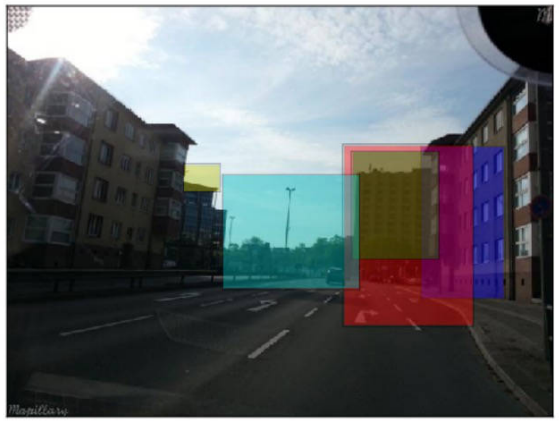
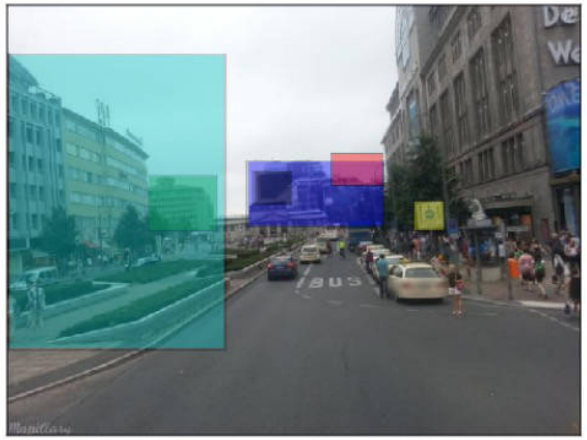
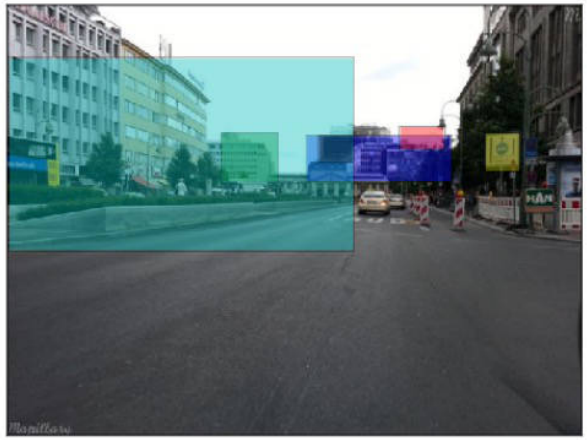
Example:
N = 1000 images
Nd = 100 words per image

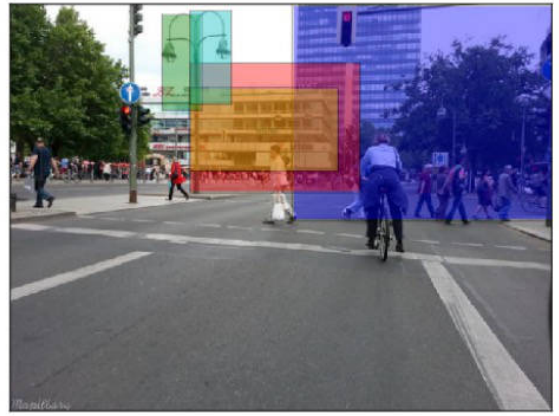
Word w1 appears 10 times in a specific image, and 100 times in all images.

Word w2 appears 10 times in a specific image, but 10,000 times in all images.

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Local Descriptors – Deep Learning



Place Recognition with ConvNet Landmarks: Viewpoint-Robust, Condition-Robust, Training-Free.
Niko Sünderhauf, Sareh Shirazi, Adam Jacobson, Feras Dayoub, Edward Pepperell, Ben Upcroft, Michael Milford. RSS 2015.

Global vs Local Image Desriptors



- one compact vector
- efficient nearest-neighbour retrieval
- may lose fine spatial information
  - → struggle with viewpoint changes



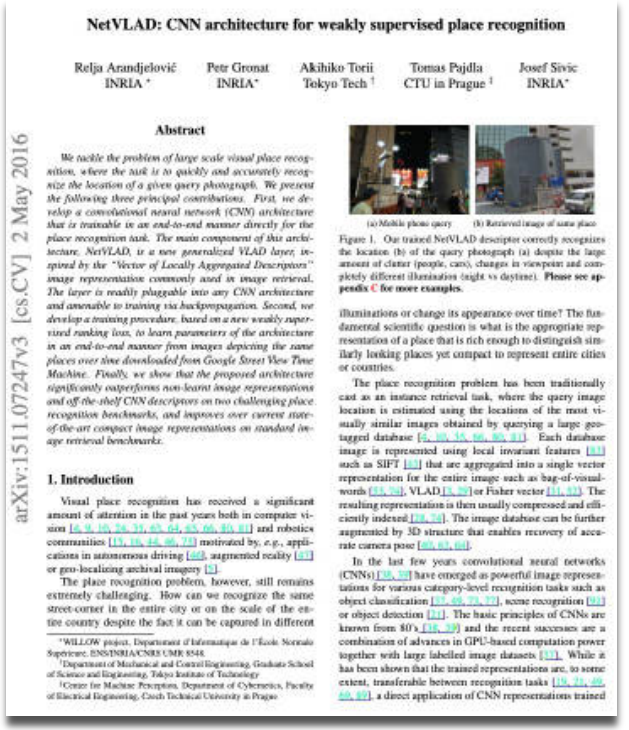
- many feature vectors
- preserve local structure
- more robust to partial overlap / viewpoint change
- more expensive to match

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Feature Aggregation

Best of both worlds:

- extract rich local features, but aggregate them into a single compact descriptor that can be indexed efficiently
- Widely used aggregation method: **NetVLAD** learns how to aggregate local CNN features into a global descriptor specifically useful for place recognition.



NetVLAD: CNN architecture for weakly supervised place recognition

Relja Arandjelović INRIA * Petr Gronat INRIA * Akihiko Torii Tokyo Tech † Tomas Pajdla CTU in Prague † Josef Sivic INRIA *

Abstract

We tackle the problem of large scale visual place recognition, where the task is to quickly and accurately recognize the location of a given query photograph. We present the following three principal contributions. First, we develop a convolutional neural network (CNN) architecture that is trainable in an end-to-end manner directly for the place recognition task. The main component of this architecture, NetVLAD, is a new generalized VLAD layer, inspired by the "Vector of Locally Aggregated Descriptors" image representation commonly used in image retrieval. The layer is readily pluggable into any CNN architecture and amenable to training via backpropagation. Second, we develop a training procedure, based on a new weakly supervised ranking loss, to learn parameters of the architecture in an end-to-end manner from images depicting the same places over time downloaded from Google Street View Time Machine. Finally, we show that the proposed architecture significantly outperforms non-learnt image representations and off-the-shelf CNN descriptors on two challenging place recognition benchmarks, and improves over current state-of-the-art compact image representations on standard image retrieval benchmarks.

1. Introduction

Visual place recognition has received a significant amount of attention in the past years both in computer vision [4, 9, 10, 28, 30, 45, 64, 65, 66, 80, 81] and robotics communities [15, 16, 44, 46, 73] motivated by, e.g., applications in autonomous driving [46], augmented reality [47] or geo-localizing archival imagery [2].

The place recognition problem, however, still remains extremely challenging. How can we recognize the same street-corner in the entire city or on the scale of the entire country despite the fact it can be captured in different

(a) Mobile phone query (b) Retrieved image of same place

Figure 1. Our trained NetVLAD descriptor correctly recognizes the location (b) of the query photograph (a) despite the large amount of clutter (people, cars), changes in viewpoint and completely different illumination (night vs daytime). **Please see appendix C for more examples.**

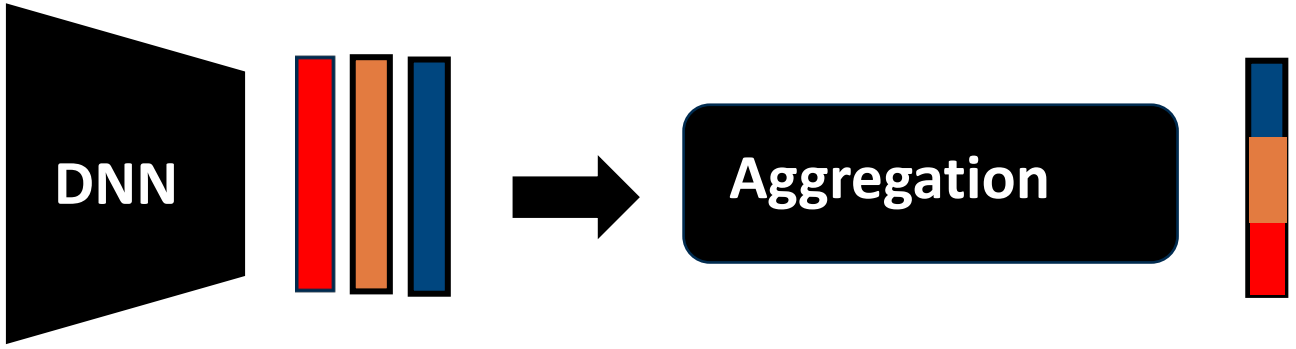
illuminations or change its appearance over time? The fundamental scientific question is what is the appropriate representation of a place that is rich enough to distinguish similarly looking places yet compact to represent entire cities or countries.

The place recognition problem has been traditionally cast as an instance retrieval task, where the query image location is estimated using the locations of the most visually similar images obtained by querying a large geo-tagged database [4, 10, 35, 66, 80, 81]. Each database image is represented using local invariant features [41] such as SIFT [41] that are aggregated into a single vector representation for the entire image such as bag-of-visual-words [53, 58], VLAD [3, 29] or Fisher vector [11, 42]. The resulting representation is then usually compressed and efficiently indexed [28, 74]. The image database can be further augmented by 3D structure that enables recovery of accurate camera pose [60, 62, 66].

In the last few years convolutional neural networks (CNNs) [39, 40] have emerged as powerful image representations for various category-level recognition tasks such as object classification [37, 69, 70, 72], scene recognition [82] or object detection [21]. The basic principles of CNNs are known from 80's [38, 39] and the recent successes are a combination of advances in GPU-based computation power together with large labelled image datasets [13]. While it has been shown that the trained representations are, to some extent, transferable between recognition tasks [19, 21, 48, 68, 69], a direct application of CNN representations trained

arXiv:1511.07247v3 [cs.CV] 2 May 2016





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Modern VPR Pipeline

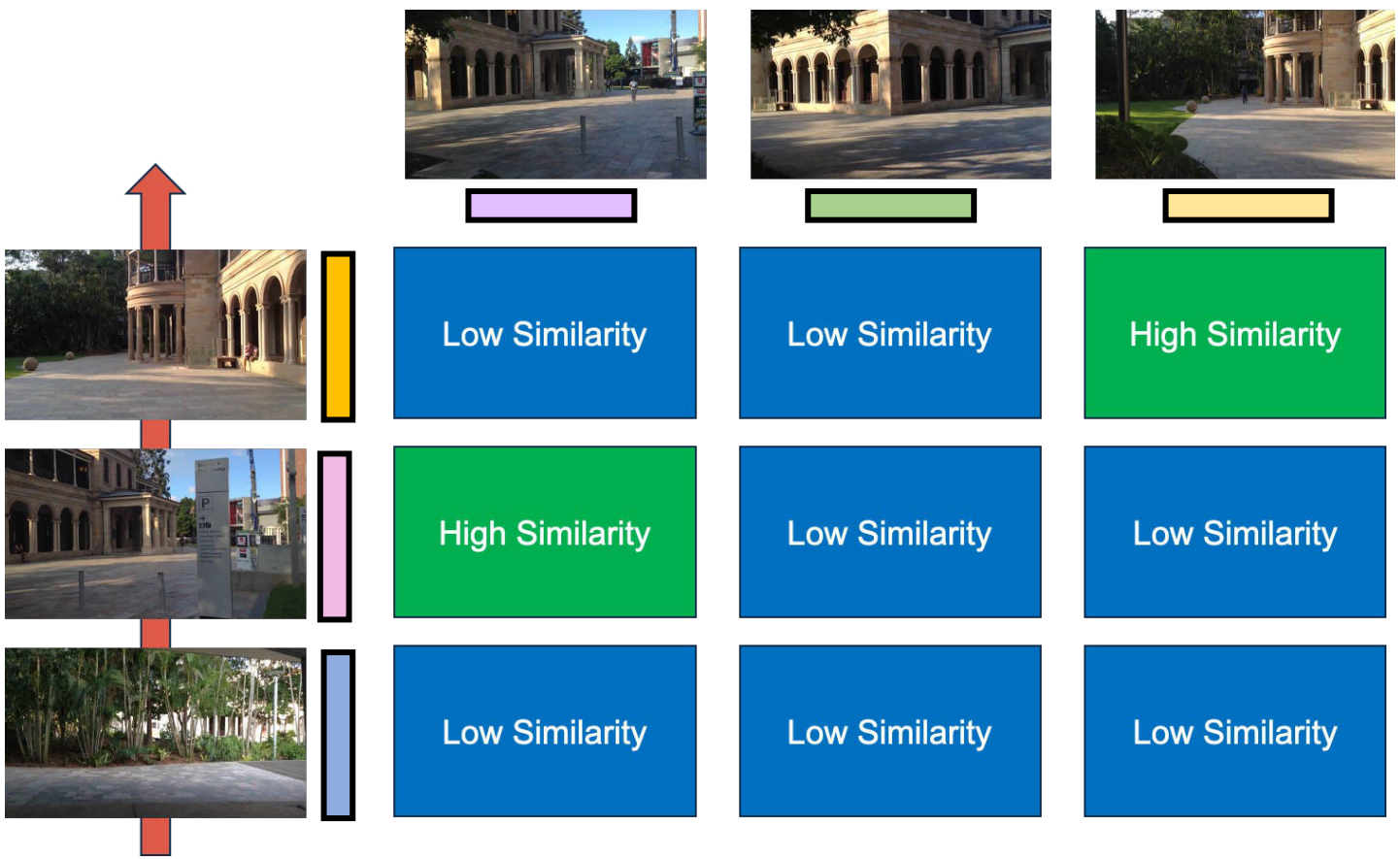
- Recent methods increasingly use large pretrained visual representations rather than learning the entire visual backbone from scratch.
- See https://anyloc.github.io/ for an example of a modern method

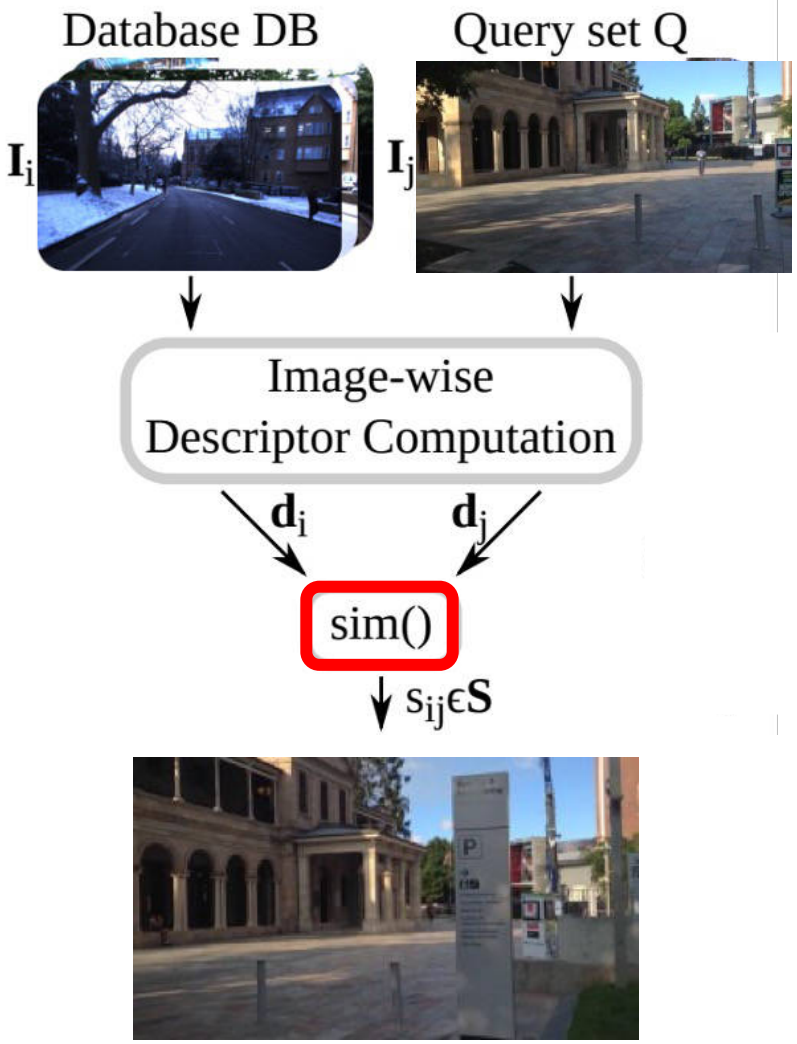


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How to learn meaningful descriptors for VPR

We need feature descriptors with high similarity for similar images





We need feature descriptors with high similarity for similar images

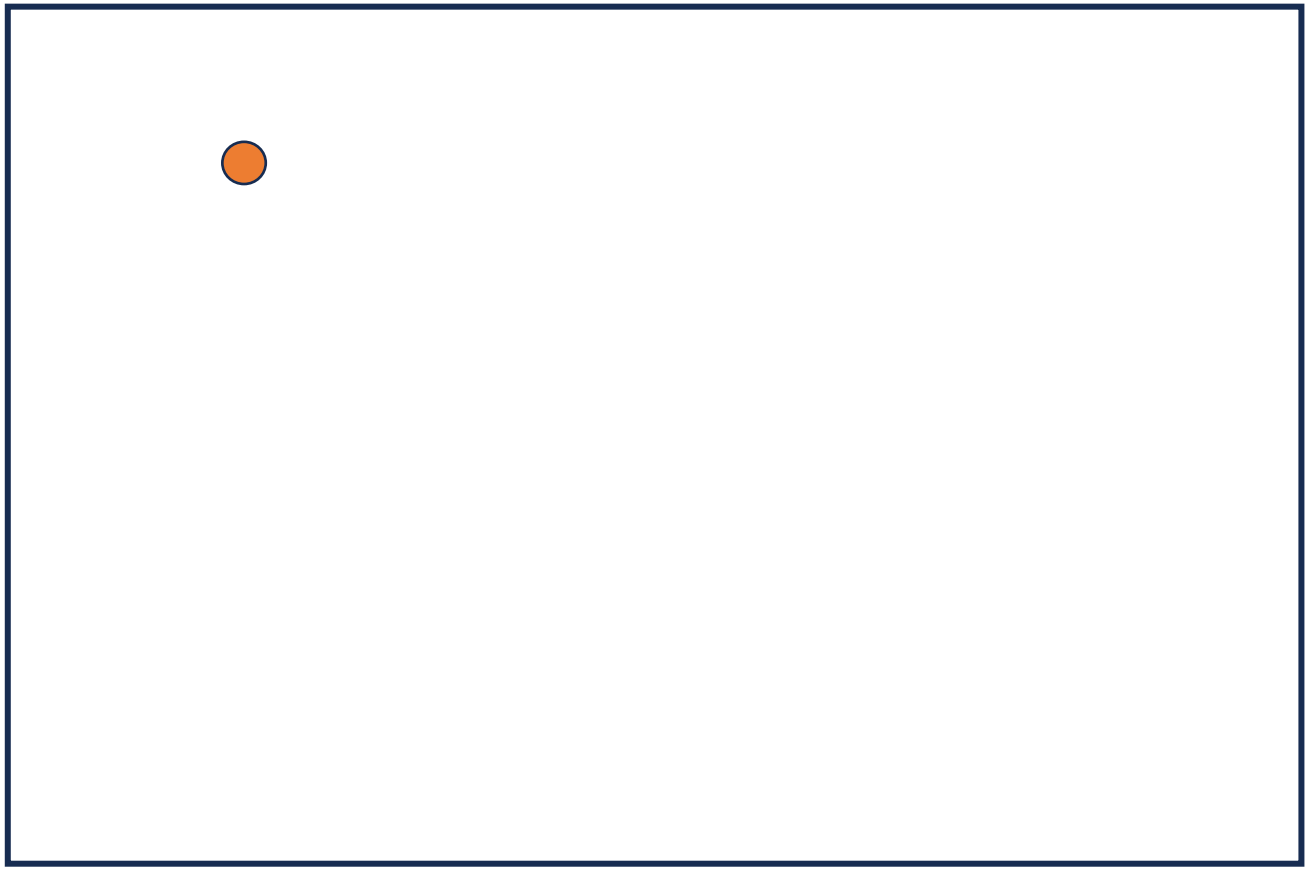


DNN

64896-d

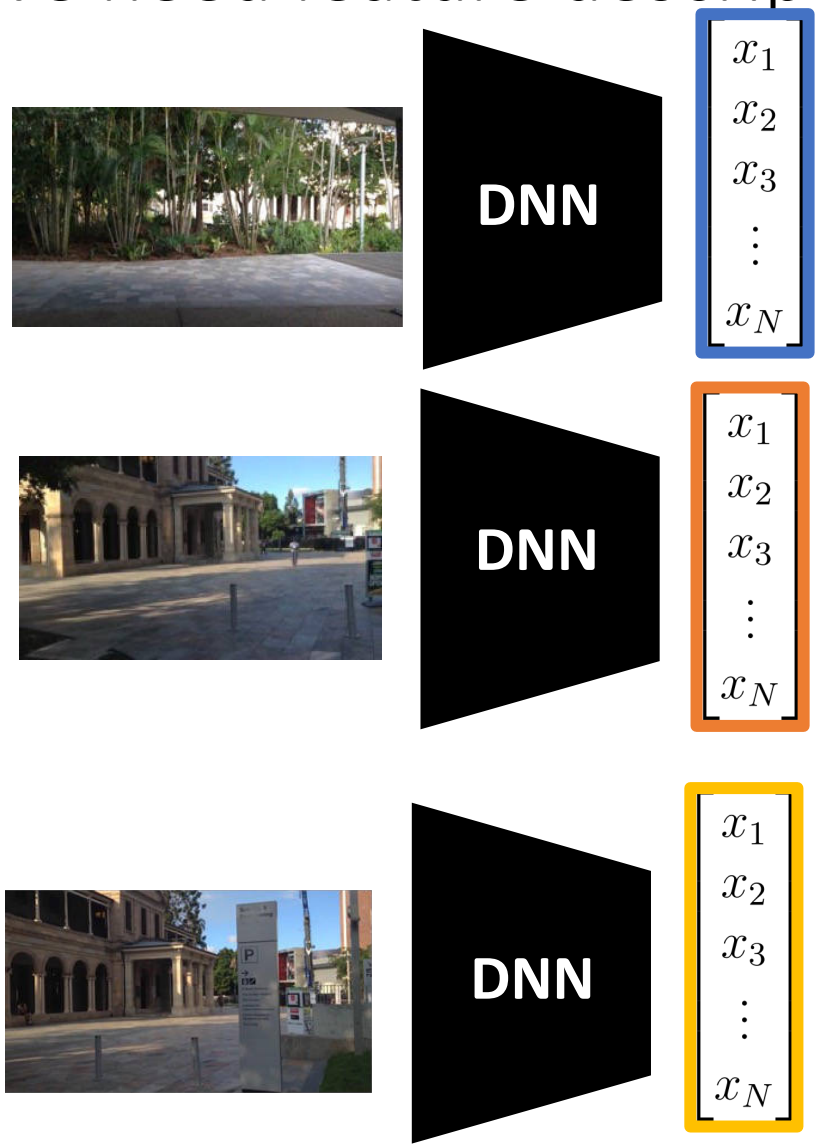
$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_N \end{bmatrix}$$

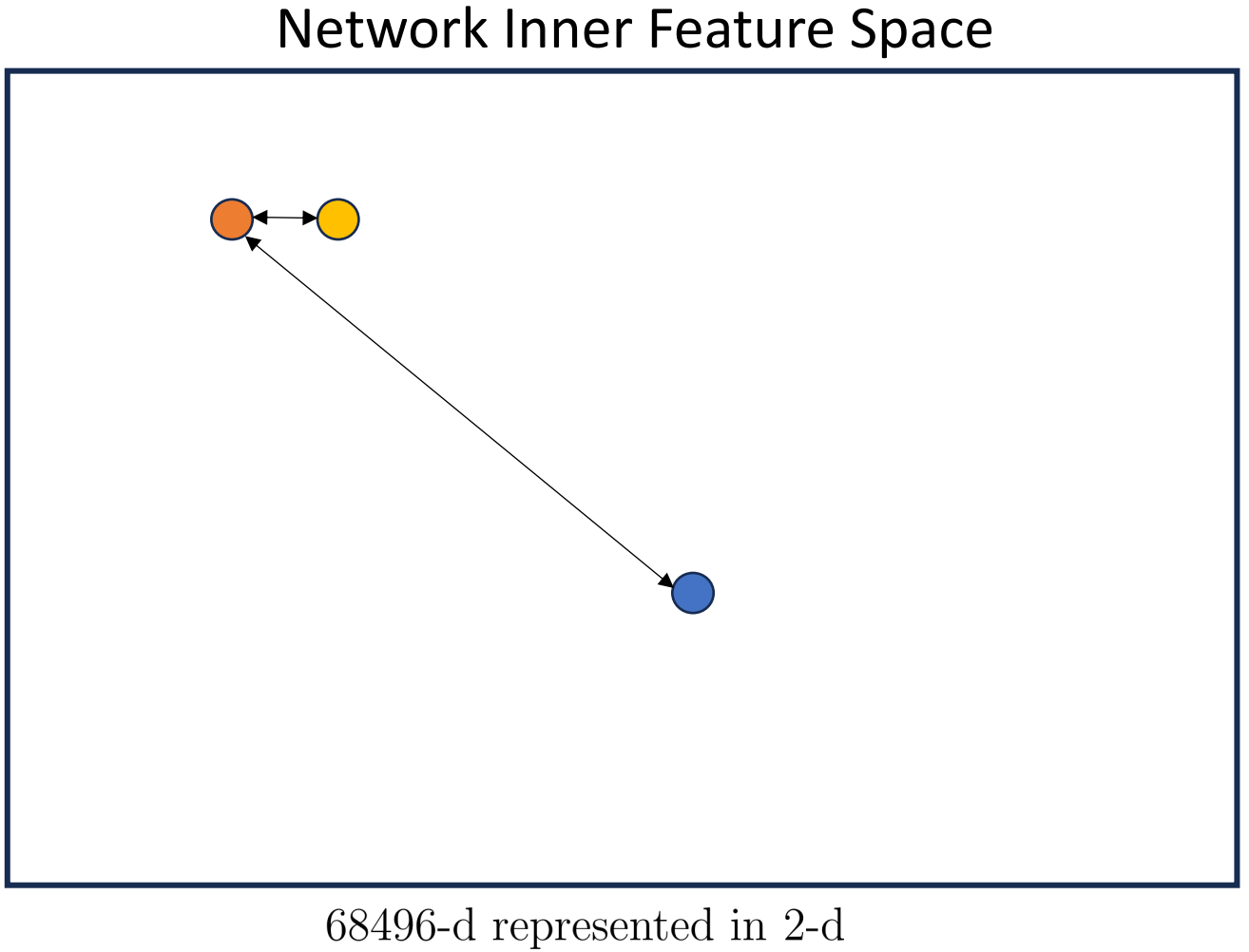
Network Inner Feature Space



68496-d represented in 2-d

We need feature descriptors with high similarity for similar images





How do we get useful VPR descriptors?

Several approaches are possible:

- Use a **pretrained representation** (without VPR-specific training)
  - ImageNet-trained CNNs historically, modern self-supervised / foundation models such as DINO

- **Metric / contrastive learning**
  - explicitly optimise distances in descriptor space
  - same-place images → close
  - different-place images → far apart
  - examples: contrastive loss, triplet loss, InfoNCE-style objectives

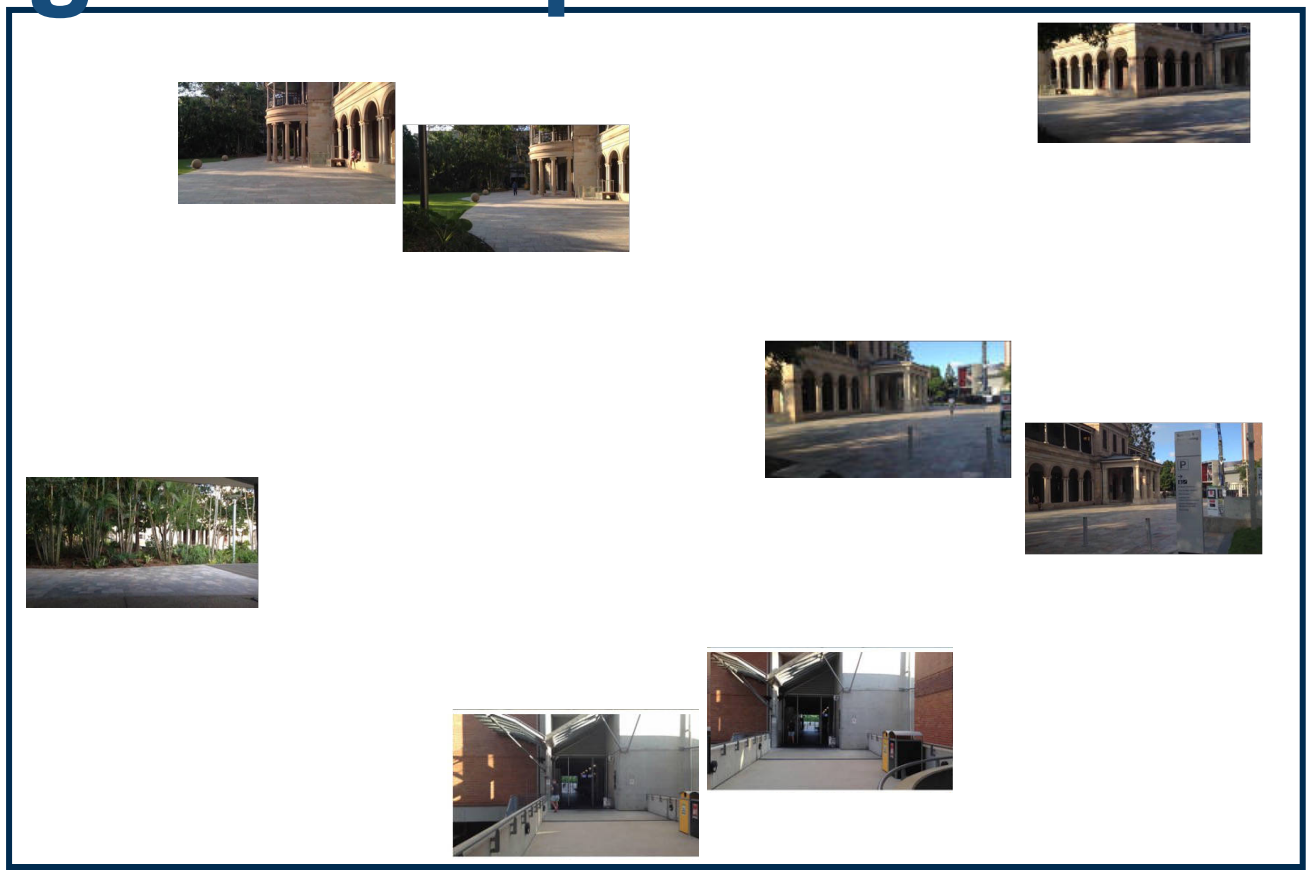
- Train specifically using place classification
  - treat locations/regions as classes → classification objective encourages discriminative representations
  - e.g. CosPlace-style approaches

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Example: Contrastive Learning with Triplet Loss

- Train a network to generate representations that place images of the same place close together

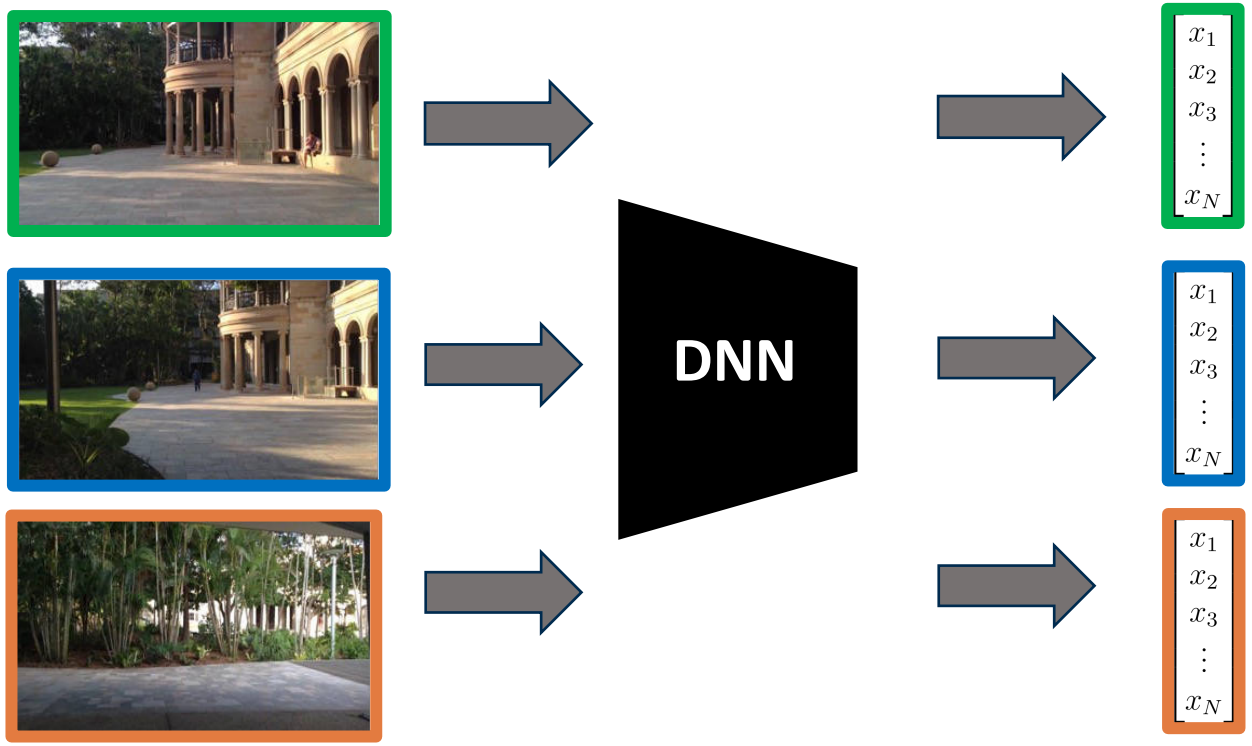
- For training: create “positive” pairs and “negative” pairs (i.e. similar and dissimilar)



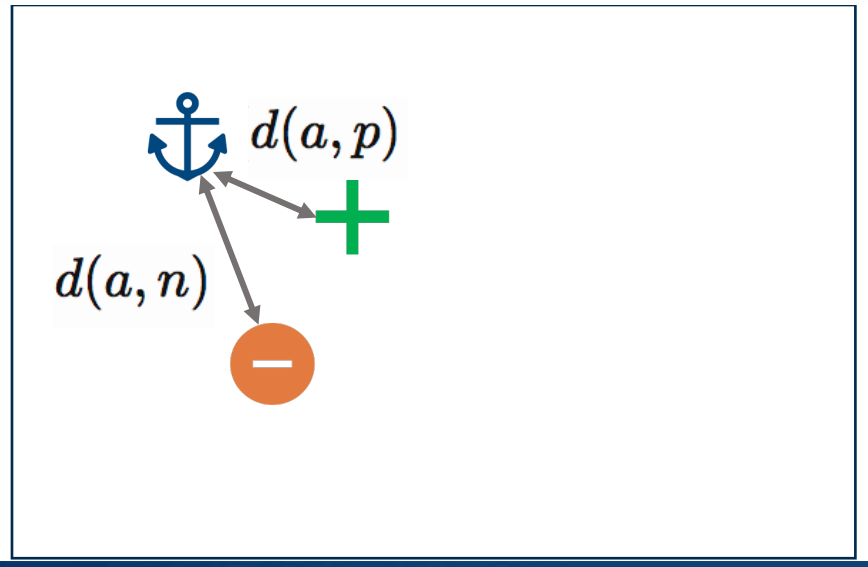
68496-d represented in 2-d

Example: Contrastive Learning with Triplet Loss

- Use a loss function that pulls positive pairs together and pushes negative pairs apart.



$$\mathcal{L} = max(d(a,p) - d(a,n) + margin, 0)$$



Contrastive Learning for VPR: Challenges

- Selecting negative samples – not too easy and not too hard
  For place recognition – How far do you move before it is a "new" place?
- GPS/geographic distance often defines positives/negatives
- viewpoint means two images can be geographically close but look very different
- hard-negative mining matters because nearby/lookalike places are the informative examples




vs.



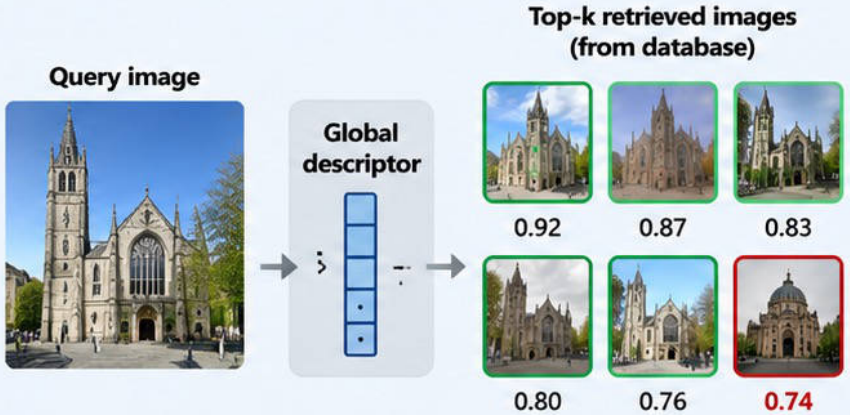

Summary and Closing Remarks

VPR is the First Step

Image retrieval gives us plausible candidates. Geometry tells us whether the match is actually valid.

1 Retrieve likely matches

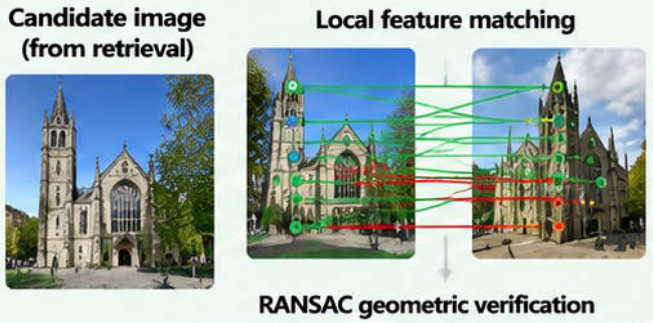
- Compute a global descriptor for the query image.
- Compare against descriptors in a large database.
- Retrieve the top-*k* most similar images.



Retrieval is fast and can search very large databases using compact global descriptors.

2 Verify the candidates

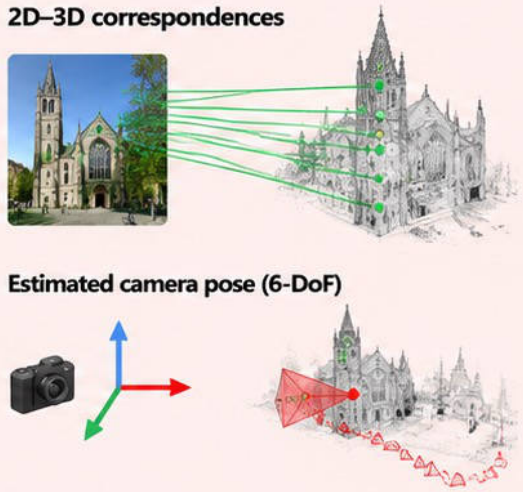
- Match local features between the query and each candidate image.
- Use geometric consistency to check if the match is valid (e.g. RANSAC).
- Reject perceptual false matches.



Geometrically consistent match
(inliers support a valid camera geometry)

3 Localise precisely

- If the candidate image is linked to a 3D map, establish 2D–3D correspondences.
- Estimate the camera pose using PnP.
- Refine the pose if necessary.



Fast but coarse

- Efficient nearest-neighbour search
- Can handle large databases (thousands/millions)
- Returns a small set of plausible candidates

High similarity does not guarantee the same place

- Perceptual aliasing can produce convincing false matches.
- Geometric verification is essential to reject incorrect candidates.

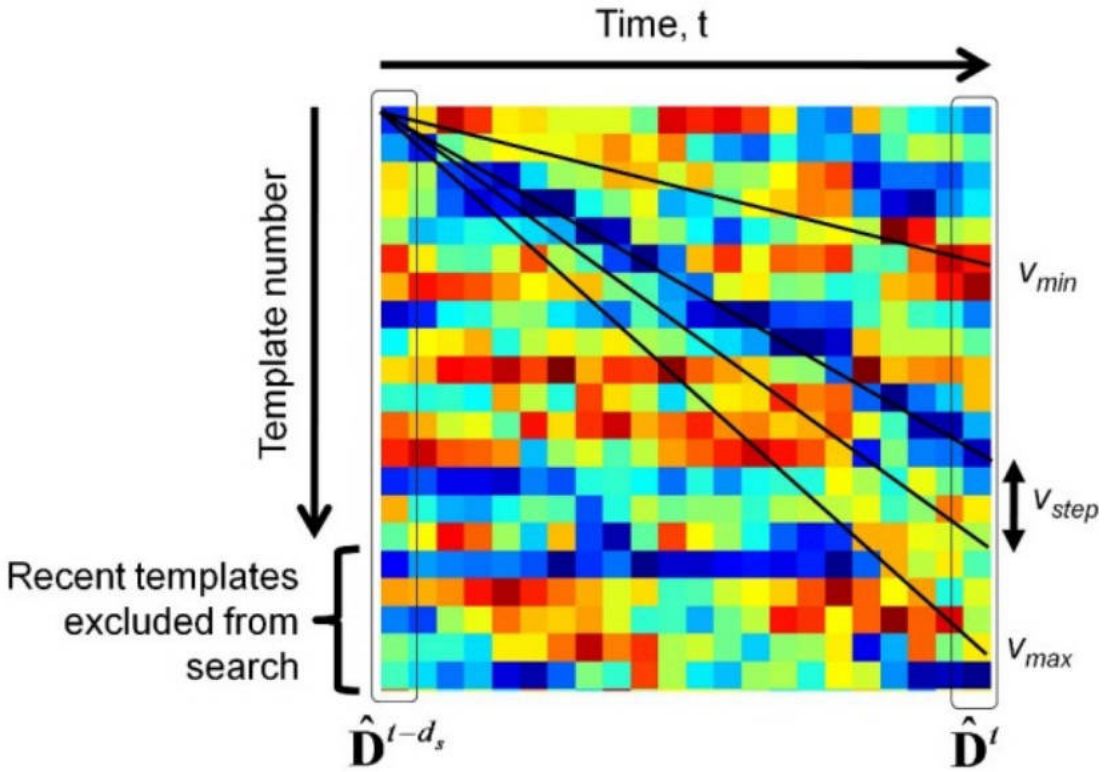
Accurate but expensive

- Local feature matching and geometry are computationally expensive.
- Only run on a small number of candidates.
- Provides a precise 6-DoF camera pose.

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Sequence-based Methods

- Instead of matching single images, try to match sequences of images
- Exploit temporal and spatial structure of the problem.
- This is one difference between VPR and Image Retrieval, VPR can make use of temporal coherence between images.



SeqSLAM: Visual Route-Based Navigation for Sunny Summer Days and Stormy Winter Nights. Milford & Wyeth, ICRA 2012.

Choose Best Method for your Application

- Questions to consider:
  - Do you require invariance against
    - Viewpoint, Illumination, Appearance, Dynamic objects, etc?
  - What is more important: high precision or high recall?
  - Online or offline (batch) mode?
  - Runtime and compute requirements

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Summary And Further Reading

- VPR is an image-retrieval problem: represent each image, compare descriptors, and retrieve likely matches.
- Modern systems commonly combine strong learned visual representations with compact global descriptors for fast retrieval.
- Retrieval is often only candidate generation — geometric verification or sequence information can make the final decision more reliable.

- Visual Place Recognition: A Tutorial. Schubert et al. IEEE Robotics and Automation Magazine. 2023
- VPR-bench: An open-source visual place recognition evaluation framework with quantifiable viewpoint and appearance change. Zaffar et al., International Journal of Computer Vision, 2021.
- Check work by QUT researchers Michael Milford, Tobias Fischer, and their PhD students and postdocs on Google Scholar.

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